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**STRATEG AI: Autonomous Strategic Planning Assistant for  
SMBs**

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## Declaration

I hereby declare that this project report is based on my original work except for citations and quotations which have been duly acknowledged. I also declare that it has not been previously and concurrently submitted for any other degree or award at UIT University or other institutions.

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# **STRATEG AI: Autonomous Strategic Planning Assistant for SMBs**

## **ABSTRACT**

With the current data-based economy, small and medium-sized firms find it difficult to make wise decisions about sourcing and pricing since they have less access to market analytics and competitive intelligence tools. This paper also deals with that challenge by creating an AI-based decision support platform that will automate data collection, data analysis, and strategic recommendations of e-commerce operations. The system is a fusion of intelligent web scraping, machine learning, and management control-based reasoning, thus converting raw market data to actionable insights.

The solution offered presents a light, modular, and easy architecture that is to process product data obtained by wholesale and retail, detect profitable opportunities, and suggest the optimal selling strategies. The platform forecasts and makes predictions, unlike the traditional dashboards or CRMs, which can only visualize data, making it bridge the gap between the presence of data and strategic decision making. Its combination with autonomous multi-agent allows conducting continuous analysis, assessing confidence, and creating an adaptive strategy, which is both scalable and future-proof.

The project highlights the role of currently available AI and automation technologies in streamlining the complex business processes and making intelligent decision-making a possibility to everyone. It is innovative because it is capable of combining technical efficiency and strategic foresight that enable businesses to compete with a lot of confidence in dynamic digital markets. The future is to expand the platform to incorporate real-time retraining pipelines and more integration with the cloud-based analytics to improve accuracy, flexibility and worldwide coverage.

**Keywords:** Artificial Intelligence, Data Analytics, E-Commerce Strategy, Business Decision Support, Automated Market Intelligence, Machine Learning.

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# 1.Introduction

## 1.1 Background of Study

The e-commerce space generates a massive volume of both structured and unstructured market data, supplier quotes, competitor prices and consumer demand data. Although such data can be translated into insights that can be leveraged upon through the use of sophisticated Business Intelligence (BI) tools in big businesses, small and medium-sized businesses (SMEs) often do not have the means or the expertise to do so.

The most recent studies on automated data extraction (Hattig et al., 2024; Huang et al., 2025) and dynamic pricing (Gupta, 2024; Zheng et al., 2024) suggest that AI and automation can leverage market data to inform business strategies. Unfortunately, many of these systems are too expensive and resource-heavy for SMEs.

To address this issue, this project aims to develop a lightweight, Agentic AI-driven decision support system that can autonomously collect, assess, and integrate market data to recommend optimal pricing and sourcing strategies.

| Problem Area    | Existing Limitation        | Impact on SMEs                  |
|-----------------|----------------------------|---------------------------------|
| Data Extraction | Manual or static scraping  | Delayed updates & inconsistency |
| Price Analysis  | Human-driven or rule-based | Poor profit estimation          |
| Market Strategy | Reactive planning          | Loss of competitiveness         |
| Tool Cost       | Proprietary BI platforms   | Financially inaccessible        |
| Problem Area    | Existing Limitation        | Impact on SMEs                  |

*Table 1.1Key Challenges in Current E-Commerce Decision Processes*

## 1.2 Problem Statement:

SMEs struggle with making timely decisions, due to the fragmented market information and their limitations and analytics. Traditional BI dashboards do not tell businesses to only

display data and do not tell businesses what strategies they can take. On the other hand, enterprise offerings are highly skilled and costly. To provide decision support that is both intelligent and affordable for small businesses, this project aims to develop an autonomous multi-agent system which can acquire data, predict real-time prices and come up with strategies to facilitate smaller firms with intelligent and affordable decision support.

## **1.3 Motivation**

Research shows that systems with multiple agents and machine learning analytics may significantly improve efficiency and profit (Roja et al., 2023; Qasem et al., 2023). However, SMEs are rarely served by such innovations. The goal of this project to democratize AI driven strategy tools so that even non-technical business owners are able to complete it. The system aims to translate complicated data analytics into basic, feasible suggestions by integrating AI, automation, and management control ideas into a one, user friendly platform.

## **1.4 Aim and Objectives**

### **1.4.1 Aim:**

To create and implement an operation, an agentic support system for decisions driven by AI that automates data extraction, analysis, and strategic recommendations for e-commerce operations that have been.

### **1.4.2 Objectives**

- To automate web scraping from wholesale and retail sources.
- To clean and store collected data in an integrated E-Commerce Database (ECDB).
- To employ machine-learning models for pricing and profit-margin prediction.
- To implement a Management-Control-Based (MCB) decision agent to evaluate confidence and formulate strategies.
- To incorporate a Marketing Agent that generates promotional and budgeting strategies.
- To present insights through an interactive dashboard for human approval.
- To periodically retrain models offline using historical ECDB data.

## 1.5 Significance of the Project and Agentic AI Approach

The project utilises an Agentic AI paradigm, in which some independent agents collaborate to accomplish particular goals. The primary objective of the system is for each agent to sense its surroundings, assess data, and act to maximize profitability through well-informed choices.

Below is a list of agents and their roles in this system

| Name                    | Agent Function                                                         | Type               |
|-------------------------|------------------------------------------------------------------------|--------------------|
| Scraper Agent           | To extract product data from wholesale and retail platforms            | Reactive Agent     |
| Analytics/Pricing Agent | To predict the optimal selling price and profit margin                 | Cognitive Agent    |
| Competitor Agent        | To identify rival pricing and promotions                               | Analytical Agent   |
| Marketing Agent         | To suggest advertising, budgeting and campaign strategies              | Planning Agent     |
| MCB Decision Agent      | To integrate outputs, check confidence, and formulate a final strategy | Deliberative Agent |
| Dashboard Agent         | To interface with user for visualization and approval                  | Interface Agent    |
| Scheduler Agent         | To trigger periodic model retraining                                   | Temporal Agent     |

*Table 1.2 Agents and their Functions*

The project differs from static analytical tools due to its agentic structure:

- **Autonomy:** Once activated, agents function without human oversight.
- **Goal Orientation:** Every agent works toward a sub-goal that is in line with the overarching company objective.
- **Coordination:** Cooperative reasoning is made possible by agents exchanging data via ECDB.

- **Adaptivity:** The system can independently re-scrape or retrain after assessing confidence.
- **Human Collaboration:** Users continue to be involved for approval and supervision.

Since intelligence goes beyond prediction to include reasoning, planning, and action, the project thus exemplifies a real-world application of agentic AI.

## 1.6 System Overview

### 1.6.1 Functional Description

The system begins by the user typing the name of the product or brand specification. The Scraper Agent would retrieve product information off wholesale and retail websites. It gathers supplier price information, minimum order information, delivery terms and retail prices of competitors.

This data is then purged to be made clean and normalized in the E-Commerce Database (ECDB) to ensure accuracy and consistency among various sources.

Upon the completion of the data flow, the Analytics Agent performs competitor analysis and pricing inference analysis, striving to come up with the average and minimum prices, evaluating the demand tendencies, and revealing potential profit margins.

These insights are then communicated to the Marketing Agent by the Analytics Agent, who comes up with marketing strategies on the basis of analytics, such as promotional offers, the medium of marketing to be employed, the intensity of the campaign, the budget and the level of profitability among others.

Management-Control-Based (MCB) Decision Agent integrates all the recommendations related to the pricing, competitors actions, and marketing projections. It estimates confidence intervals, determines whether the suggested profitability levels have been met and sets up an optimal plan of action.

Finally, the Dashboard Agent will show all the recommendations, optimal selling price, and profit margin and market actions to the user in an interface. The user can approve, edit, or request re-analysis.

### **1.6.2 Subsystems / Modules**

- The **Data Acquisition Module (Scraper Agent)** gathers product information from wholesale and retail websites, including prices, suppliers, and competitor details.
- The **ECDB (Data Management Module)** cleans, verifies, and organizes this data so it's ready for analysis.
- The **Analytics or AI Module (Analytics Agent)** studies competitor data, finds pricing trends, and predicts the best-selling price along with profit margins.
- Marketing Module (Marketing Agent) offers the recommendation of promotions, advertisement budgets, and campaign techniques to enhance the product visibility and sales.
- The final business strategy is formed by the Decision and Strategy Module (MCB Agent), which uses the opinion of all the other agents, implements the confidence levels, and creates the end result of the business plan.
- All the insights, pricing decisions, and the marketing plans are displayed in the easy-to-understand dashboard available to users through the Visualization Module (Dashboard Agent).
- The Offline Training Process (Scheduler Agent) periodically retrains the AI models based on stored information to ensure the results are up to date and comply with the changes in the market.



### 1.6.3 System Performance Specifications

| Parameter            | Target Value          | Description                                    |
|----------------------|-----------------------|------------------------------------------------|
| Scrape Cycle Time    | $\leq 30$ s per query | Sequential multi-source data collection        |
| Model Inference Time | $\leq 2$ s            | Real-time price C profit prediction            |
| Confidence Threshold | $\geq 0.70$           | Minimum confidence for strategy execution      |
| Model Accuracy       | $\geq 80\%$           | Validation accuracy against benchmark datasets |
| Retraining Interval  | 7–30 days             | Offline background model update                |

*Table 1.3 System Performance Metrics and Targets*

## **2. Background and Literature Review**

### **2.1 Introduction**

The research, frameworks, and industrial systems that support intelligent e-commerce decision-making are reviewed in this chapter. We synthesize evidence from five pillars: (i) product matching and automated data acquisition; (ii) dynamic pricing and profit optimization; (iii) multi-agent decision support (Agentic AI); (iv) marketing analytics and recommendation engines; and (v) commercial solutions that reflect the state of practice at the moment. While determining the remaining gaps this project fills, we aim to map how these strands come together to form a single, reasonably priced architecture appropriate for SMEs.

### **2.2 Background**

#### **2.2.1 Data abundance, SME constraints**

The shift to e-commerce has introduced a market where handling information and applying it in a useful way is what makes a business effective. Sellers on the internet, service providers, and online marketplaces generate a huge pool of raw data such as product listings, pricing, description and reviews. Similar data can be used by firms to make better decisions by analyzing the information retrieved from platforms in a more efficient and faster way. Tools are spread-out through the market to assist businesses to bloom, but SMEs are the underdog when it comes to the use of these complex systems. This is caused by either the budget, lack of domain educated and skilled agents, or not having access to these high end tailored systems [1] [2].

#### **2.2.2 Evolution of AI in E-Commerce**

Artificial intelligences is one of the biggest growing factors for businesses to assist in decision making in context to the online economy. Earlier, online firms used spreadsheets which were manual (and still do), or use rule-based dashboards to get useful information from the ingested data. Machine learning's evolution, natural language

processing (NLPs) has made predictive analysis possible, with extra features such as demand forecasting and price optimization [3]. This work had been dominated by special, highly-paid teams which had to go through intense studies and iterations to come up with a suggestion. In recent times, multi- agent systems (MAS) and Agentic AI systems are being popularized to help break down complex problems for businesses into smaller modules handled by automated agents which interact with each other under a combined shell. [4]

### 2.2.3 From Descriptive to Prescriptive Analytics

These are upgrades in analytics, which provide prescriptive and predictive analysis by analyzing past events, and suggesting what could be the best step to take in the future. As a product, a dynamic pricing algorithm can update prices in real time (affected by the competitors and vendors), with the automated scrapper running as the core to normalize and standardize market data [6] [7]. However, an all-in-one combined solution is still not accessible to SMEs which can analyze data and give predictive analysis in order to guide decisions.

### 2.2.4 SME pain points that persist

Four persistent hurdles impede SME adoption: (1) **cost/infrastructure** of enterprise tools [2], (2) **fragmented data** across sites/APIs [9], (3) **limited ML/cloud expertise**, and (4) **reactive cycles** driven by after-the-fact reporting. Bridging this gap demands a low-cost, modular, and autonomous stack that merges data acquisition, analytics, and strategy into one continuous loop.

### 2.2.5 Emergence of Agentic AI

The intersection of machine learning, autonomous agents, and reasoning systems with continuous perception, decision, and action is known as agentic AI [10]. The agents work semi-independently accomplishing specialized tasks (i.e. scraping, pricing,

marketing, or coordination) and all communicate by sharing a common knowledge base. The method has scalability, fault tolerance, and configurability of monolithic AI models [11]. Studies have indicated that in complex environments that involve parallel decision making like dynamic pricing and supply-chain management, agentic systems are capable of doing better than single-model architectures [12]. The architecture is based on the concept where many agents working together in this project, carry out the data acquisition, data analysis, and strategic recommendation.

These developments demonstrate the increasing adult nature of single AI methods but they are still sporadic and this highlights the requirement of holistic, low-cost, and flexible multi-agent systems as suggested in this project.

### **2.2.6 Automated Data Acquisition & Product Matching**

Large-scale product corpora and benchmarks make entity matching and schema harmonization measurable at web scale. The WDC Product Corpus v2 aggregates ~26M offers and 79k websites for entity-matching research [10]. Transformers such as Ditto encode attributes for pairwise matching [11], while multimodal models (text + image) improve accuracy by 10–20% on ambiguous listings [12]. Cost-efficient modular sieve pipelines triage pages into lightweight rules and call heavier extractors only when needed, cutting compute by ~60–70% [9].

This implies that a hybrid pipeline is most practical for SMEs — use Playwright + BeautifulSoup for robust scraping, apply string/TF-IDF filters to prune, then run pre-trained transformer embeddings for fine matching, finalizing in a schema-aware ECDB. This retains much of the accuracy of neural matchers while avoiding GPU budgets.

### 2.2.7 Dynamic Pricing and Profit Optimization

Traditional methods (cost-plus, elasticity regression) are simple but inflexible in volatile markets. Supervised ML (Random Forest, XGBoost) handles non-linearities and interactions in features like MOQs, supplier cost, shipping time, and competitor averages — with excellent accuracy/cost trade-offs for SMEs [16], [17]. Reinforcement learning (DQN, DDQN, PPO) treats pricing as sequential decisions and shows 10–25% profit gains over rule-based baselines in controlled studies [5]–[7].

**Its implications is** pair XGBoost for real-time price prediction with a lightweight control loop to manage confidence and re-analysis. This captures the adaptivity of RL without the heavy data/compute burden.

### 2.2.8 Multi-Agent Decision Support (Agentic AI)

Multi-Agent Systems (MAS) distribute specialized roles (scraping, analytics, marketing, coordination) across autonomous agents that collaborate via shared state [31], [32]. Empirical studies report speedups and better resilience in market contexts where uncertainty and concurrency dominate [13], [14]. Agentic AI extends MAS with planning and reflective behaviors — agents can set subgoals, evaluate outcomes, and modify strategies [15], [33], [34].

## 2.3 Comparative Review of Related Platforms

### 2.3.1 Strategic and Analytical Interfaces.

Many AI-based strategy solutions have been created in recent times that offer automated recommendations based on sales and financial information. ProfiMind AI (2024) provides textual summaries of business activity based on large language models (LLMs) and does not have processes to collect and integrate data directly or into the market [8], also Watson

Orchestrate by IBM makes the workflow automation available as some agents, but it is more enterprise-focused [9].

While such LLM-based tools demonstrate the potential of generative strategy assistants, their closed and costly nature limits accessibility for SMEs

### **2.3.2 Dynamic Price Engines**

AI internet retail has been focused on dynamic pricing. The automated pricing engine at Amazon constantly runs prices of products in pre-set ranges depending on the competitive data and stock indications [10]. The Mastercard Dynamic Yield system utilizes real-time content and pricing recommendations to users based on reinforcement learning (RL) [11]. Both systems succeed, but are stuck in proprietary systems.

Related scientific research also proposes the application of reinforcement learning in maximizing revenues. Gupta and Kaur [12] used Deep Q - Q-Networks (DQN) to control online prices, whereas Zheng et al. [13] used Double DQN to stabilize learning. Although RL-based approaches achieve superior revenue optimization, they demand high computing and extensive data, restricting adoption by SMEs.

### **2.3.3 Customer Marketing Intelligence and Analytics.**

Multi-Agent Systems (MAS), coordinate various positions (scraping, analytics, marketing, coordination) among autonomous agents, which communicates through shared state [31], [32]. In the market settings characterised by uncertainty and concurrency, observational studies observe accelerations and enhance reliability [13], [14]. Agentic AI builds upon MAS and brings in planning and reflective capabilities - agents are capable of defining subgoals and calculating consequences and altering strategies [15], [33], [34].

Implication: A distributed, data-centric data-centric MAS interconnected by an ECDB is ideally suited to e- commerce, and allows parallel scraping, simultaneous analytics, and coordinated strategy-building under a layer of management control.

#### **2.3.4 Business Intelligence Dashboards.**

Analytical software Zoho Analytics and HubSpot Marketing Hub is available to SMEs with data visualization and reporting tools [18] [19]. These dashboards show performance measures in terms of sales growth and conversion rates but do not have independent analytical functions. They are based on manual data input and predictive modelling and automated decision making is not supported.

Altogether, the table below shows the bright examples of commercial systems that represent a larger picture of the AI-driven commerce.

### **3. Aim and Statement of Problem**

#### **3.1 Introduction**

In today's digital economy, companies collect a vast amount of data, both structured and unstructured, daily. This includes information such as competitor prices, supplier lists, customer feedback, and market trends. Large organizations use advanced AI platforms and business-intelligence suites to handle this large amount of data, but small and medium-sized businesses (SMEs) often don't have the knowledge, tools or money to use these tools properly.

The issue transcends technology; it concerns accessibility, integration, and decision-making autonomy. Most tools on the market only do one thing well, like visualization or price tracking. They don't offer a full intelligence cycle that collects, analyzes, interprets and suggests strategic actions in real time.

This project fills that gap by creating and putting into use a multi-agent, AI-based decision-support system that brings together automated web scrapping, product matching, predictive pricing, market optimization, and management-control mechanisms into a single system.

#### **3.2 Problem Definition**

AI has changed the way people do business online, but only big companies, like Salesforce Einstein, Dynamic Yield, ProfitMind AI, that can afford complicated exclusive solutions are seeing the benefits. These systems need large datasets, expensive subscriptions and IT teams that work only on them.

The main issue this project is trying to solve is the lack of an AI platform that is affordable, clear and connected, and that can independently gather, analyze, and suggest market strategies for small and medium-sized enterprises.



Some core problems are:

- The users have to manually combine different systems because current tools require manual integration.
- There is a lack of clarity in AI-generated decisions.
- The infrastructure and licensing costs are very high.
- These tools are not flexible enough to adjust when new market data or changes in data structure appear.

The suggested solution offers a single, agentic system that executes human-like strategic reasoning in data collection, pricing analysis, and marketing planning. This allows for automation that is clear and cost-effective.

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### **3.3 Aim of the Project**

The primary aim of the project is to make a lightweight, multi-agent, AI-powered business strategy assistant that can gather, analyze, and interpret e-commerce data on its own to give small and medium-sized businesses (SMEs) advice on pricing, marketing and strategy.

Specific objectives are:

1. To use modular and LLM-enhanced scraping to automate the gathering and standardization of data on products and competitors.
2. To use XGBoost and Random Forest regressors to make predictive models for analyzing profit margins and prices.
3. To make marketing and ROI forecasting tools that turn analytical results into

business plans that can be put into action.

4. To bring agents together in a single architecture that makes sure of openness, clarity, and decision validation through confidence scoring.
5. To use an affordable and open-source stack to deploy the system so that small and medium- sized businesses can use it.

### **3.4 Problem Context and Stakeholders**

#### **3.4.1 Business Stakeholders (SMEs)**

The problem here lies in reliance on manual pricing and marketing decisions because there are no integrated AI tools. There is a need for affordable platform that merges analytics, strategy generation, and visualization, the benefit to this is direct access to real-time intelligence that boosts competitiveness.

#### **3.4.2 Technical Stakeholders (Developers & Researchers)**

The challenge is to combine diverse models and data sources into a single architecture. There is a need for scalable APIs and a modular system design. This project showcases an effective application of agentic AI in real-world SME settings.

#### **3.4.3 End Users (Marketers & Decision-Makers)**

There is trouble in trusting unreliable models or interpreting analytics outputs. There is a need for clear, comprehensible suggestions with assurance levels. The system enables users to make informed pricing and marketing decisions with greater confidence and transparency by offering data-driven insights through dashboards.

### **3.5 Scope of the Project**

#### **In Scope:**

- Web-scraping from multiple wholesale and retail platforms will be done in real-time.
- The data collected will be cleaned and organized using ECDB.
- Price and profit-margin predictions will be made using supervised learning

techniques.

- Linear regression will be used to generate recommendations for marketing strategy and ROI.
- A confidence-based MCB loop will handle decision validation.

**Out of Scope:**

- Supply chain management, logistics, and payment system integration are not included.
- Large-scale RL training that requires cloud GPUs is also excluded.
- Future work will involve real-time deployment across several e-commerce sites.

### **3.6 Distinctive Features and Competitive Edge**

- **A single Agentic AI architecture** that combines all the essential business operations into a single collaborative AI framework, including pricing, marketing, analysis, and scraping.
- **SME centric design** that reduces financial barriers by using open-source software and free-tier cloud tools.
- **Explainability and Transparency** are ensured through agents recording decisions and confidence levels to enable human oversight.
- **Scalability** is also ensured by integrating new agents or models without redesigning the system, due to the modular design.
- **Technology Independence** is also ensured; unlike proprietary vendor-locked systems, the system can run locally or on cloud platforms.

### 3.7 Technical Problems and Solutions

| Area                   | Problem Encountered                             | Solution / Implementation                                                    |
|------------------------|-------------------------------------------------|------------------------------------------------------------------------------|
| <b>Data Collection</b> | Site structures are frequently changed          | We do modular scraping with LLM fallback for adaptive extraction.            |
| <b>Data Cleaning</b>   | Product schemas are heterogeneous               | ECDB normalization based on schemas will be done.                            |
| <b>Model Training</b>  | Absence of labeled data                         | Semi-supervised learning and publicly available benchmark datasets are used. |
| <b>Integration</b>     | Synchronization between agents                  | For communication, we used the ECDB shared-memory approach.                  |
| <b>User Interface</b>  | Visualization of the outputs of multiple agents | Flutter UI integrated with a Power BI dashboard.                             |
| <b>Deployment Cost</b> | Budget is limited                               | Hosted entirely on free-tier platforms (Supabase, Render, Vercel).           |

*Table 3.1 Identified Challenges and Implemented Solutions During System Development*

## **4. Hardware, Software analysis and requirements**

### **4.1 Introduction**

This chapter covers the technical foundation of StrategaAI. It compares various hardware and software options, describes the methods used for requirement discovery, and provides justification for the ultimate technological decisions. As the solution operates as a Platform as a Service (PaaS) deployed in the cloud, it depends on software orchestration rather than dedicated hardware.

To provide adaptive decision intelligence, the system integrates Next.js, LangGraph + CrewAI agents, Supabase + Redis data layer, and FastAPI.

### **4.2 Fact-Finding Techniques**

#### **4.2.1 Literature review:**

It helped identify the research gaps and trustworthy AI techniques of dynamic pricing and multi-agent systems by reviewing academic articles and for-profit tools such as Salesforce Einstein, ProfitMind AI, and Dynamic Yield.

#### **4.2.2 Stakeholder Surveys:**

It helped in learning about the key performance indicators, usability expectations, and price optimization challenges of SME owners, marketers, and analysts.

#### **4.2.3 Competitive Analysis:**

To assess the shortcomings of the current pricing and marketing tools, a competitive analysis was carried out using benchmarking platforms such as Reprice.io, Shopify Collabs, and the Amazon Repricing Engine.

#### **4.2.4 Observation & Prototype Testing:**

Streamlit-based prototypes were used for observation and prototype testing in order to verify the dashboard and user interface requirements, guaranteeing their clarity and usability.

### 4.2.5 Data sampling:

It was performed to understand the diversity of data sources by collecting supplier and retail listings from platforms such as Alibaba and Made-in-China.

## 4.3 Hardware Analysis

### 4.3.1 Hardware Environment

| Component                                    | Purpose                        | Specification                                |
|----------------------------------------------|--------------------------------|----------------------------------------------|
| Developer Machine                            | Model training & testing       | Intel i7 CPU, 16 GB RAM, 512 GB SSD          |
| Cloud Hosting (Vercel + Render + Supabase)** | Deployment & database services | 2 vCPUs, 2 GB RAM (free-tier)                |
| End-User Devices                             | Access dashboards              | Any modern browser device ( $\geq$ 4 GB RAM) |

*Table 4.1 Hardware Environment*

### 4.3.2 Why Software Not Hardware

The platform uses computational logic rather than physical sensing since it is manipulating digital data, like text, prices, and reviews. Therefore, intelligence exists in the form of software agents, APIs, and machine learning models. Hardware acts only as a host.

## 4.4 Software Analysis

### 4.4.1 Frontend Layer:

We chose **Next.js** (TypeScript) + **TailwindCSS**

| Alternative                 | Pros                                                     | Cons                       |
|-----------------------------|----------------------------------------------------------|----------------------------|
| React.js                    | Mature ecosystem                                         | Manual routing, no SSR     |
| Angular                     | Strong MVC framework                                     | Heavy for SaaS             |
| Flutter Web                 | Cross-platform                                           | Poor SEO & web integration |
| <b>Next.js<br/>(Chosen)</b> | Server-side rendering,<br>TypeScript, easy Vercel deploy | Slight learning curve      |

*Table 4.2 Comparison of Frontend Framework Alternatives*

### Justification:

Next.js gives us server-side rendering for SEO and performance. TailwindCSS makes design fast and responsive. They create together a clean low latency UI ideal environment for real time dashboards

### 4.4.2 Backend Layer

We chose **FastAPI**

| Alternative             | Reason Rejected                                                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Node.js (Express)       | It has Weak type safety and limited ML integration                                                                                                      |
| Django                  | It has heavy monolith for microservices                                                                                                                 |
| Flask                   | It gives no async support                                                                                                                               |
| <b>FastAPI (Chosen)</b> | It gives asynchronous processing, a Python-native machine learning stack, and automatic validation to ensure fast, reliable, and efficient performance. |

*Table 4.3 Comparison of Backend Framework Alternatives*

### Justification:

FastAPI integrates smoothly machine learning libraries in Python (scikit-learn, XGBoost) and supports asynchronous concurrency, which is a core for multi-agent workflows.

### Middleware / Agent Framework – LangGraph + CrewAI

| Alternative                        | Limitation                                                  |
|------------------------------------|-------------------------------------------------------------|
| LangChain                          | It has heavy runtime overhead                               |
| Haystack                           | It is suited for document QA only                           |
| <b>LangGraph + CrewAI (Chosen)</b> | It enables directed multi-agent flows and task coordination |

*Table 4.4 Comparison of Agent Framework Alternatives*

### Edge:

It supports orchestration which is based on graphs where each agent (Scraper, Analytics,

Marketing, Decision) acts partially in an autonomous way but shares state via Supabase.

### 4.4.3 Web Scraping Layer

We chose **Playwright + BeautifulSoup**

| Alternative                   | Issue                                    |
|-------------------------------|------------------------------------------|
| Scrapy                        | It is static only (no JS)                |
| Selenium                      | It is slow and resource-intensive        |
| <b>Playwright (Chosen)</b>    | It automates dynamic JS pages            |
| <b>BeautifulSoup (Chosen)</b> | Lightweight HTML parser for static pages |

*Table 4.5 Comparison of Web Scraping Tools*

#### Justification:

The hybrid combination provides speed and resilience against website structure changes.

### 4.4.4 Database / Storage

We chose **Supabase (Postgre SQL) + Redis**

| Alternative                      | Why Rejected                                                                        |
|----------------------------------|-------------------------------------------------------------------------------------|
| Firebase                         | It is proprietary and doesn't support SQL joins                                     |
| MongoDB Atlas                    | It has weak relational constraints                                                  |
| AWS RDS                          | It gives paid service                                                               |
| <b>Supabase + Redis (Chosen)</b> | It provides free-tier, real-time capabilities, built-in auth, and efficient caching |

*Table 4.6 Comparison of Database and Storage Options*

#### Edge:

Supabase Realtime API enables agents to communicate through one data stream—acting as shared “memory” for the system.

### 4.4.5 Model & AI Stack

| Algorithm | Alternative Rejected | Reason for Chosen Model |
|-----------|----------------------|-------------------------|
|-----------|----------------------|-------------------------|



|                                |                                   |                                        |
|--------------------------------|-----------------------------------|----------------------------------------|
| <b>XGBoost / Random Forest</b> | Deep RL (DQN, PPO) too data-heavy | High accuracy + explainable            |
| <b>Linear Regression</b>       | Genetic Algorithms complex        | Simple and transparent                 |
| <b>MCB Control Loop</b>        | Full RL controllers               | Lightweight confidence logic           |
| <b>TF-IDF + Transformers</b>   | Custom BERT training is expensive | Pre-trained NLP for feature extraction |

*Table 4.7 Comparison of Algorithms and Model Selection*

These algorithms maintain the system's interpretability and affordability while guaranteeing prediction accuracy.

#### 4.4.6 Visualization Layer

We chose **Power BI + Streamlit**

| <b>Tool</b>      | <b>Purpose</b>                                                               | <b>Audience</b>       |
|------------------|------------------------------------------------------------------------------|-----------------------|
| <b>Power BI</b>  | It provides executive dashboards with KPIs (ROI, Margins, Pricing Trends)    | Business Users        |
| <b>Streamlit</b> | It provides technical monitoring of model health, confidence, and retraining | Developers / Analysts |

*Table 4.8 Comparison of Visualization Tools*

#### Why Both:

Streamlit provides two complementary layers of visualization: live analytical feedback and model retraining, while Power BI provides polished, shareable business insights.

## 4.5 Algorithms and Data Structures

| Agent        | Algorithm                 | Purpose                             |
|--------------|---------------------------|-------------------------------------|
| Scraper      | Modular Sieve + RAG LLM   | For extracting data adaptively.     |
| Analytics    | XGBoost / Random Forest   | For predicting price and profit.    |
| Marketing    | Linear ROI Regression     | For budget and channel forecasting. |
| MCB Decision | Confidence Threshold Loop | To validate recommendations.        |
| Scheduler    | Prefect Cron Flow         | For periodic retraining             |

*Table 4.9 Algorithms Used by Each Agent*

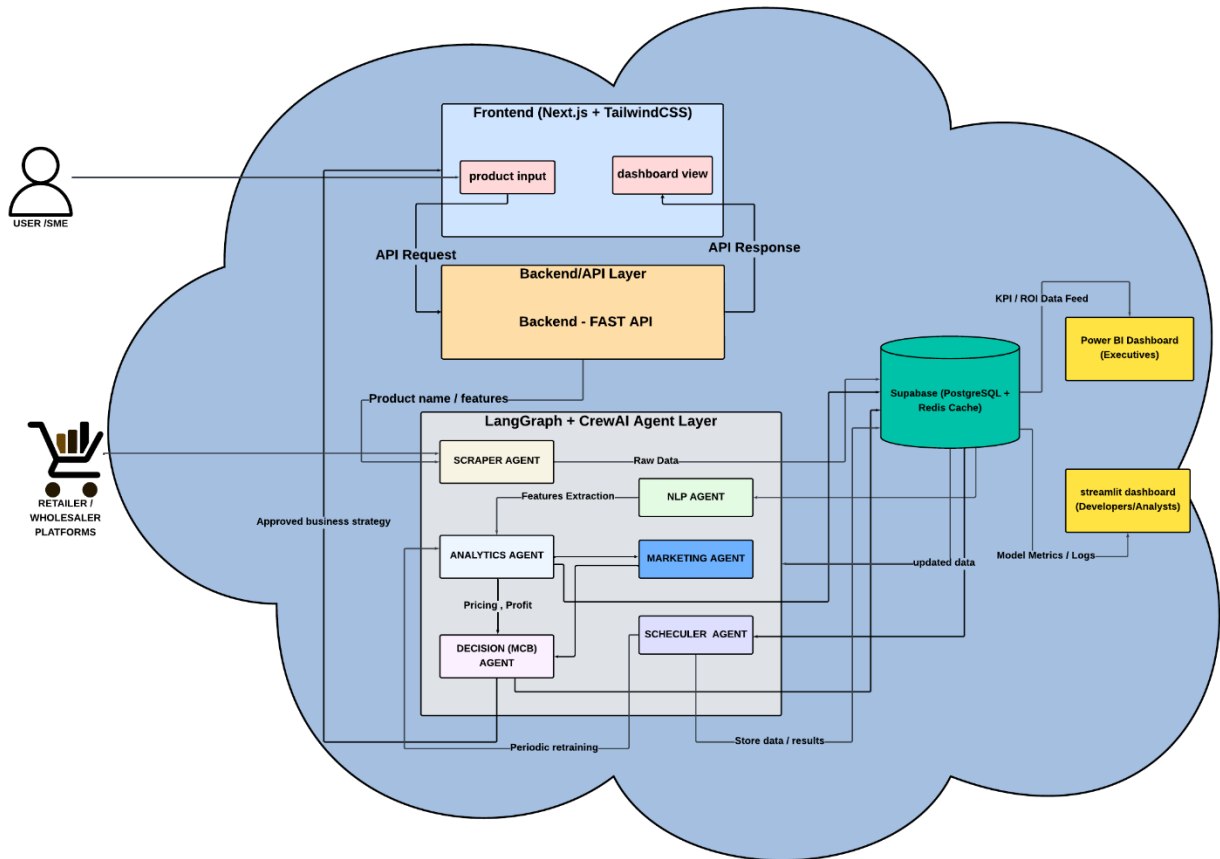
### Data Structures:

JSON objects for inter-agent messages, relational PostgreSQL tables (products, competitors, strategies), Redis key-value cache for temporary states.

## 4.6 Overall System Structure

### 4.6.1 System Architecture

This diagram illustrates the overall cloud-based deployment of the platform. It shows how the frontend (Next.js + TailwindCSS) communicates with the backend (FastAPI) and connects to multiple intelligent agents through the LangGraph + CrewAI layer. The Supabase database stores structured data, while Power BI and Streamlit dashboards visualize results for both executives and analysts.

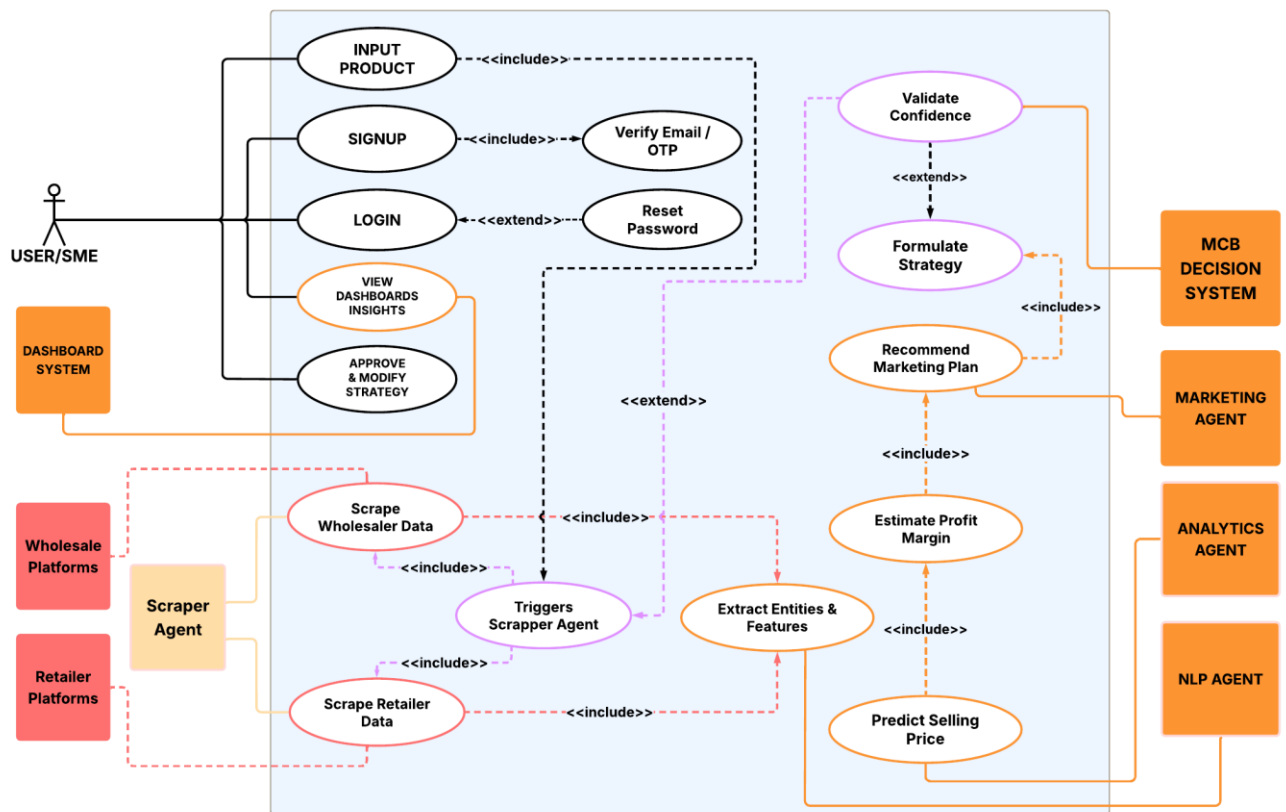


*Figure 4.1 System Diagram Of StrategAi*

## 4.6.2 Actor Use Case Diagram

**Actors:** Business User, External Commerce Platforms (as data sources).

**Modules:** Data Acquisition, Analytics, Decision & Strategy, Visualization.



*Figure 4.2 Actor Use Case Diagram of StrategAi*

## 4.6.3 Activity Diagram

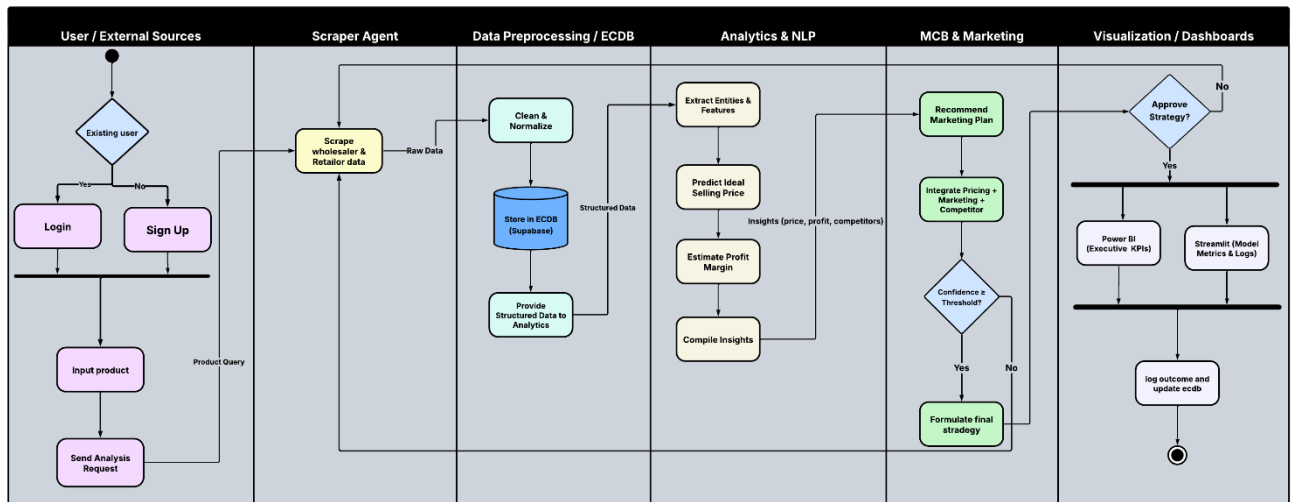


Figure 4.3 Activity Diagram of StrategAi

## 4.7 User Stories and Acceptance Criteria

### 1. Product Analysis

#### User Story:

As a business owner, I want to input a product so that I can see market pricing.

#### Acceptance Criteria:

**Given** a valid product name, **When** I start analysis, **Then** the system scrapes wholesale and retail data and updates ECDB within 60 seconds.

### 2. Price Optimization

#### User Story:

As a finance manager, I want optimal price suggestions so that I can maximize profit.

#### Acceptance Criteria:

**Given** supplier data, **When** analytics runs, **Then** it returns optimal price & margin with  $\geq 70\%$  confidence.

### 3. Marketing Plan

#### User Story:

As a marketer, I want AI ROI forecasts so that I plan budgets effectively.

#### Acceptance Criteria:

**Given** profit margins, **When** marketing agent executes, **Then** it shows ROI and recommended channels.

#### **4. Decision Validation**

##### **User Story**

As a decision-maker, I want to see confidence scores so that I can trust outputs.

##### **Acceptance Criteria:**

**Given** MCB confidence < 0.70, **When** strategy is reviewed, **Then** system re-runs analysis automatically.

#### **5. Visualization**

##### **User Story:**

As a user, I want to view insights in dashboards so that I can approve strategies.

##### **Acceptance Criteria:**

**Given** stored results, **When** I open Power BI or Streamlit, **Then** I see updated KPIs and model metrics.

## **5. Software Design and Modeling (The Complete Vision)**

### **5.1 Introduction**

StrategAI is being developed, as an advanced intelligent system that supports human decisions in rapidly changing business situations. It does not act like one big application but rather as a well-coordinated family of smart parts which keep watching, evaluating, learning, and acting back. This book gives the entire presentation of StrategAI by showing how platform architecture, agent-driven logic, and data structure merge toward enabling scalable self-directed adaptive business intelligence. The paper seeks not only to explicitly state the technical aspects but also to foreground how every design move contributes toward the cumulative building of dependability, plug-in growth, and real-time awareness.

### **5.2 System Architecture: The Agentic Core**

StrategAI does not live in the simple-tool realm, but finds its readiness within the real MADSS (Multi-Agent Decision Support System). Its source design lies at the heart of agent-enabled collaboration on an agentic basis-meaning that rather than localizing intelligence it is deliberately distributed. Thus, scraping, analytics, NLP, and marketing agents too operate as independent reasoning units with perception-interpreting-action abilities. They collaborate and coordinate by sharing what they know individually through global knowledge space which technically is collective memory-in our case. So setting know-how in various parts makes strength allows modularity and puts StrategAI ready to grow with added needs and areas.

#### **5.2.1 Macro-Architectural Vision (The 3-Tier Foundation)**

##### **Interface Tier:**

The link layer creates a two-view method, giving use at work with plan watch. A live dashboard fueled by Vite-React where users can see agent actions and checks and start tasks with near zero delay comes with Power BI or Streamlit shaped study reports for leaders able

to look deeper into trends and give choice smarts. This mix gives quickness, and picture truth, and insight find the right level for both choosers and tech users.

### **Orchestration Tier (FastAPI & Agent Framework):**

This tier represents the logical brain where the ecosystem of agents — such as the Scraper, NLP, Analytics, Marketing, and MCB agents — execute their specialized tasks. Built on FastAPI, this layer enables asynchronous processing, scalable API interaction, and efficient inter-agent communication. Concepts inspired by LangGraph introduce controlled directed cycles and state-aware task flows, ensuring that agents can loop intelligently through workflows while maintaining context. This design allows the system to handle multi-step reasoning, recover from interruptions, and coordinate complex decision processes smoothly.

### **Persistence Tier (Supabase/PostgreSQL & Redis):**

This layer represents the logical brain in which the ecosystem of agents comes into play, that is, Scraper, NLP, Analytics, Marketing, and MCB agents performing their topical tasks. In this tier, FastAPI enables asynchronous processing and scalable API interaction with efficient inter agent communication. Concepts inspired by LangGraph introduce controlled directed cycles and stateful taskflows, ensuring that agents will loop intelligently within workflows while keeping context. This design allows for multi-step reasoning in the system to be recovered upon interruption of the complex decision processes being coordinated.

## **5.3 Comprehensive Module Modeling**

The project architecture sits within a group of seven standalone working parts, purposely joined by their basic thinking role inside the bigger agent setup. StrategAI does not keep its smartness as one large block of rules but spreads it among Quick, Thoughtful, Plan-Making, and Considering helpers so that each can focus, grow, and work with the shared ECDB. Having separate parts at this stage makes sure that every part adds new



ability, also from noticing the outer shop world all the way to thinking about price plans and long-term choice rule, yet in a joined and planned manner.

### **5.3.1 Data Acquisition & Normalization (Reactive Agents)**

#### **The Scraper Agent (The Sensors):**

This agent serves as the sensory layer of the platform that continuously collects structured and unstructured information in a wide range of retail and wholesale ecosystems? in, e.g., the Daraz, Mega.pk, MadeinChina, and Alibaba websites. IT uses Modular Sieve architecture to adapt to any page layouts, APIs and any content structures in such a way that product listing, price reconfiguration, availability indicators and other significant attributes are acutely maintained. It is not accompanied with the main task of gathering raw information but has instead made available the data to the system in an incredibly uniform and highly processable form which can be used as the lowest level in all other higher levels of intelligence.

#### **The NLP Agent (The Pre-processor):**

Once the data about products has been acquired, the NLP Agent transforms this raw data into useful groups of products identities out of the noisy text. It uses such methods as TF-IDF vectorization and cosine similarity to attract shapes in titles and descriptions of products. In the process, it will be in a position to understand which listing available on different marketplaces are pointing to the same underlying commodity despite the differences in spelling or words or even branding. Through this normalization process, StrategAI can regard the same products as one record and not as part records to a given downstream level intelligence modules.

### **5.3.2 Intelligence & Prediction (Cognitive Agents)**

#### **The Analytics/Pricing Agent (The Mind):**

It is placed on the central part, therefore regarded as a thinking agent and uses deep predictive

modeling to get the best market pricing strategies. It is built on the ensemble machine-learning architecture (poorly put, XGBoost using Random Forest) that performs a synthesis of competitor density signals in history on profit margin and MOQ restrictions. This is then translated into price suggestions that would not only make it continue to make profits but also maintain its competitiveness in the market. The rationale is more than responsive to the market action and takes the proactive action of the probable possibilities and therefore permits an informed strategic pricing choice.

#### **The Competitor Agent:**

Following the pricing logic, this special analyzing mod follows competitor moves. it picks up promo breaks, price reductions and "stock- Out" tips - indicating when competitors lose supply advantage or adopt aggressive price gambit. These discoveries assist StrategAI to direct prompt solutions such as price reduction or advertisement drives in such a manner that sellers may retain or increase market portion whenever a chance of competition is on the anvil.

### **5.3.3 Strategic Planning & Strategy Suggestion (Planning Agents)**

#### **The Marketing Agent (Strategic Engine):**

This agent is an auto CMO of STP and analytics sees the product-level dynamics of the STP framework. It determines which side fits more with the bulk Divorcing B2B or retail consumers B2C through information relative to price variation and MOQ pattern recommending target audience to apply it at the time can make a positioning plan to determine whether or not the product is competing as Value, Premium or Budget offering depending on competitor prices, quality clues and market margin prospects.

Based on this, the agent creates a tactical 4Ps mix:

- **Product:** It suggests bundle pricing - "Buy2 and save 10%" where low MOQ can be used to generate sales-volume based on scale, etc.

- **Price:** Collaboration with the Analytics Agent to gain an advantage of a market promoting, but margin-optimal pricing structure.
- **Place:** Recommending the most appropriate channel selection which is either Daraz Flash Sales that will generate fast-moving sales or Facebook retargeting that will build the brand in the long-term.
- **Promotion:** Formulates promotion concepts, seeks areas of promotion, and recommends advertising budget constraints to aid in execution.

#### **The MCB Decision Agent (The Governor):**

This agent does serve as the evaluative evaluation filter since it acts as the last reviewer such that only plausible, valid and justifiable strategies are made available to the user. In a system under the MCB thought, such agent gives weight (confidence score between 0 and 1) to all other agents on the advice to give. A certain point of confidence (say 0.70 and higher) at which only strategies that outperform it are admitted are permitted. This prevents non-certain analytics, noisy data or riskiness into the system; therefore, responsible system practice under executed strategic judgment.

### **5.3.4 Maintenance & Evolution (Temporal Agents)**

#### **The Scheduler Agent:**

The Scheduler Agent will make sure that StrategAI is constantly changing and evolving so that it does not become stagnant or irrelevant. It does periodic background retraining cycles which are usually 7-30 days long and during this, the ML models update themselves with the current historical trends in the ECDB. This aspect works to reduce the direction of the models so that predictions and findings ensure conformity to the present market facts and not to the historical facts. By doing so, StrategAI will act like a living system where it never stops learning and updating its intelligence in the long run.

## 5.4 UML Diagrams

### 5.4.1 Class Diagram

This diagram represents the final state of the pure project, showing how future modules (Marketing, MCB) integrate into the base system.

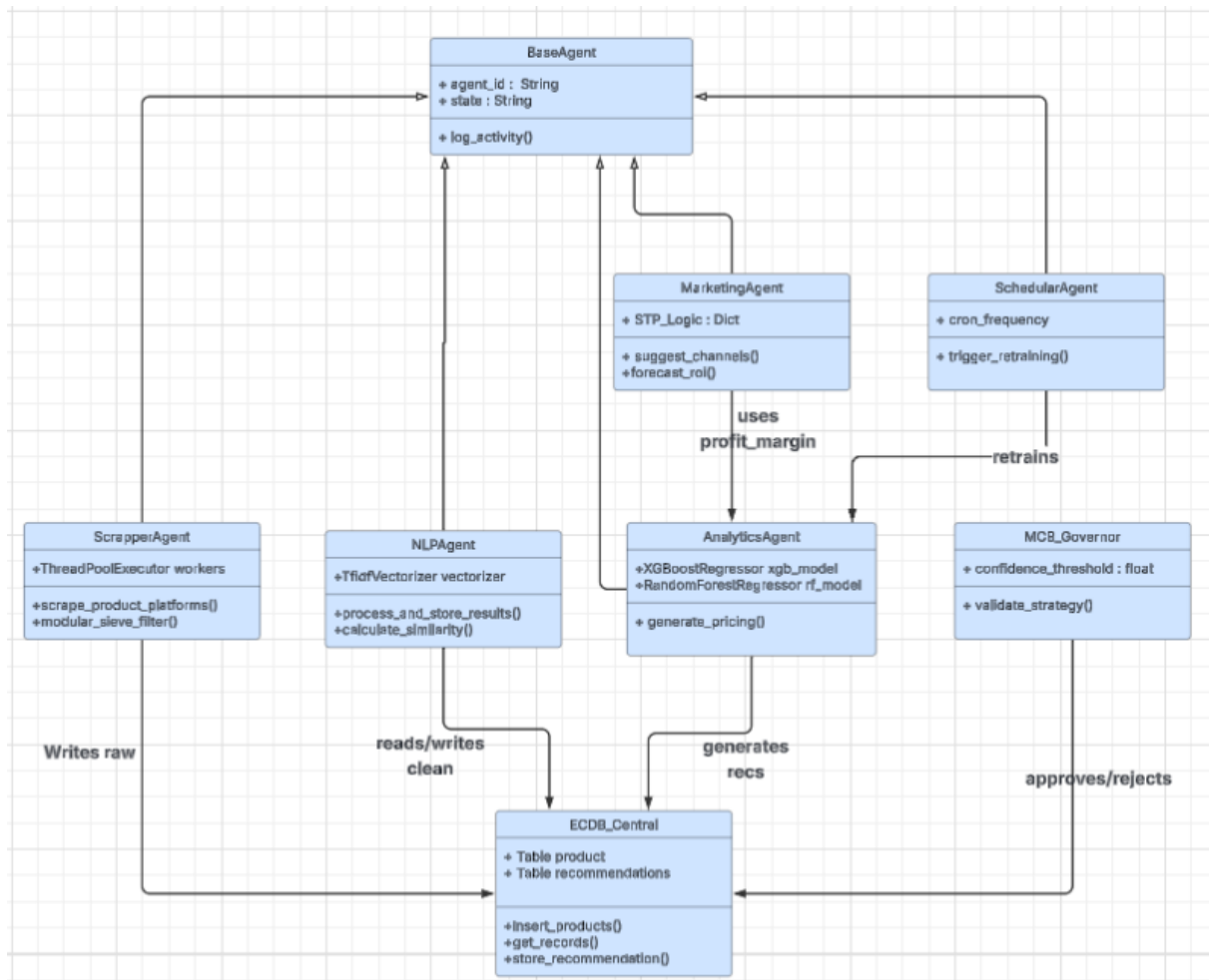
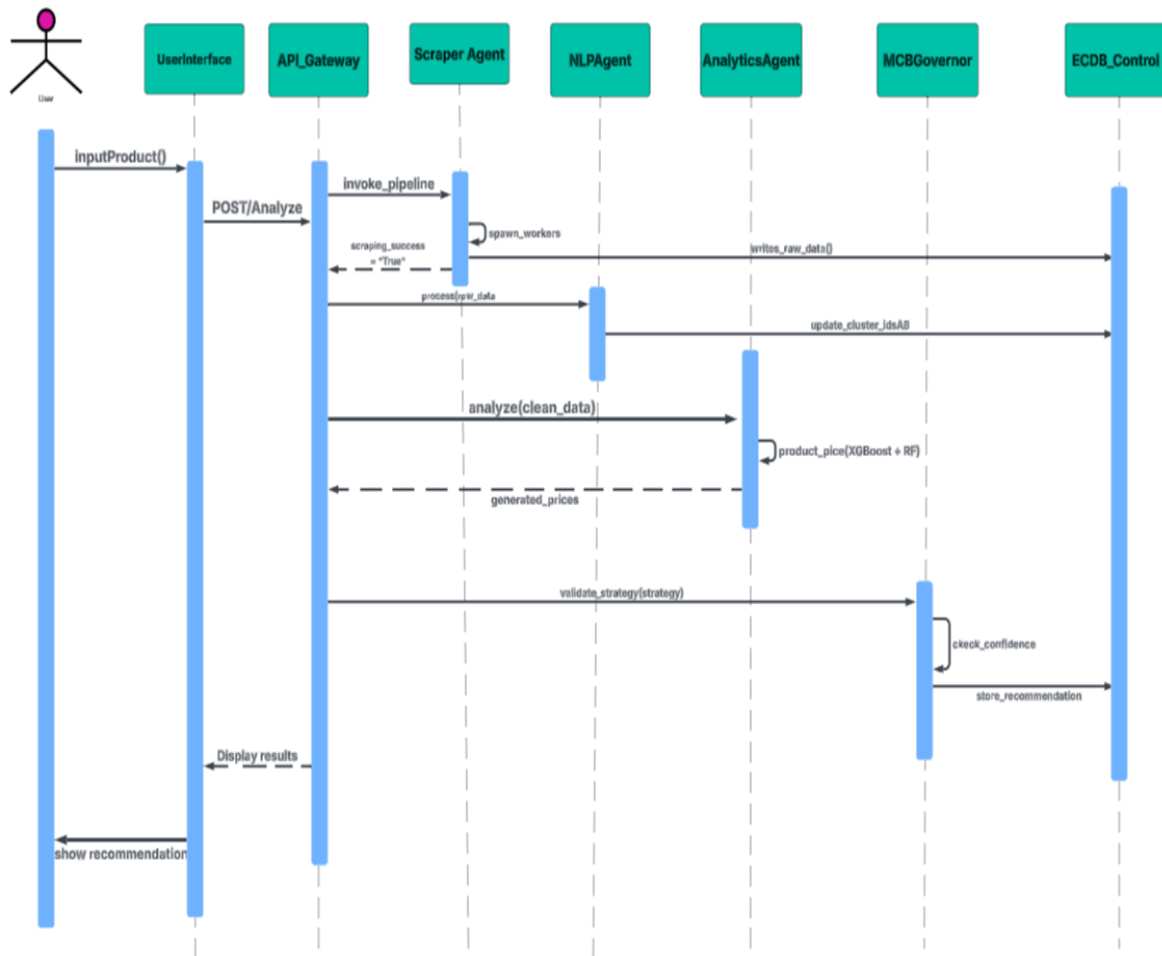


Figure 5.1 Class Diagram of StrategAi

### 5.4.2 The Multi-Agent Interaction Sequence

Modeling the "Intelligent Relay Race" from User Search to Marketing ROI.



*Figure 5.2 Sequence Diagram of StrategAi*

This sequence diagram shows how a user's product search moves through the system like a relay race. The user enters a product, the API Gateway sends the request to the Scraper Agent to collect and clean the data, and the NLP and Analytics agents analyze it using machine-learning models to generate prices and insights. The strategy is then validated and stored, and finally the system returns a clear recommendation back to the user.

### 5.4.2.1 Sequence Diagram of Scrapping Sequence

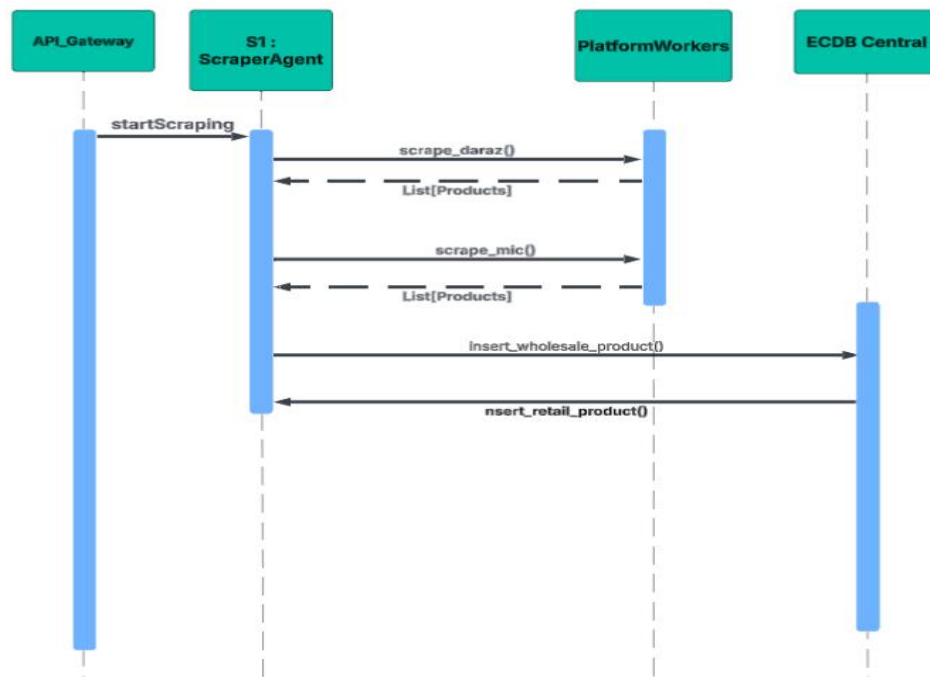


Figure 5.3 Sequence diagram of Scrapping Sequence

### 5.4.2.2 Sequence Diagram of NLP Cleaning Sequence

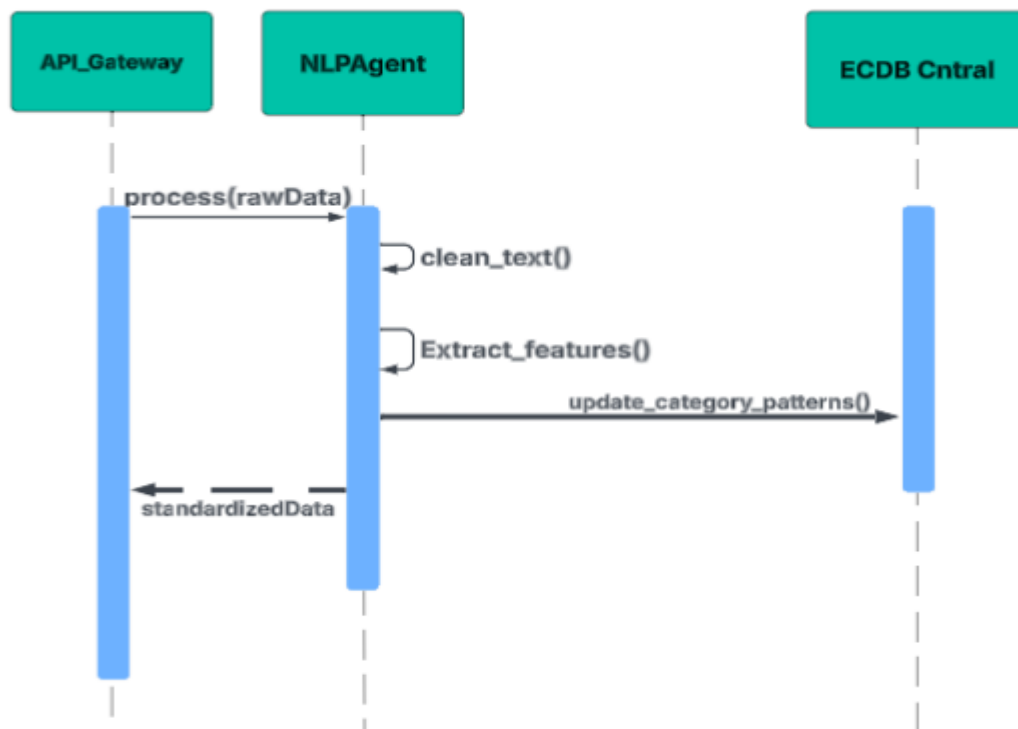
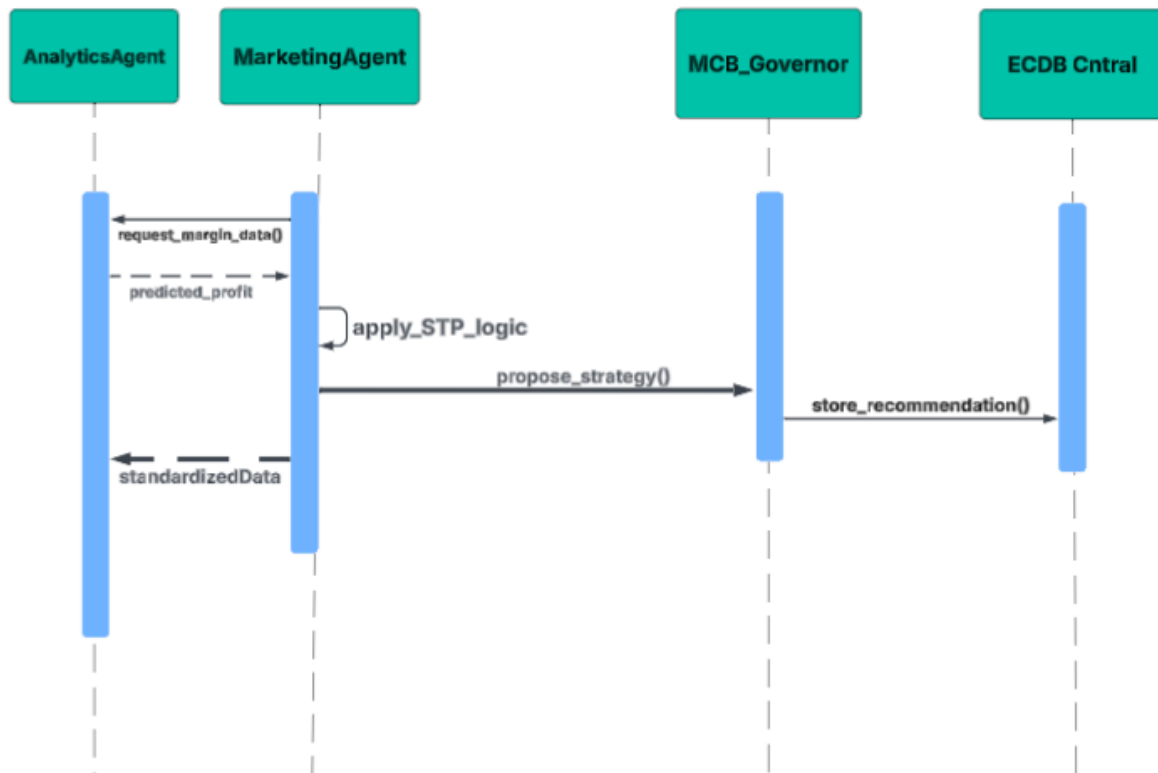
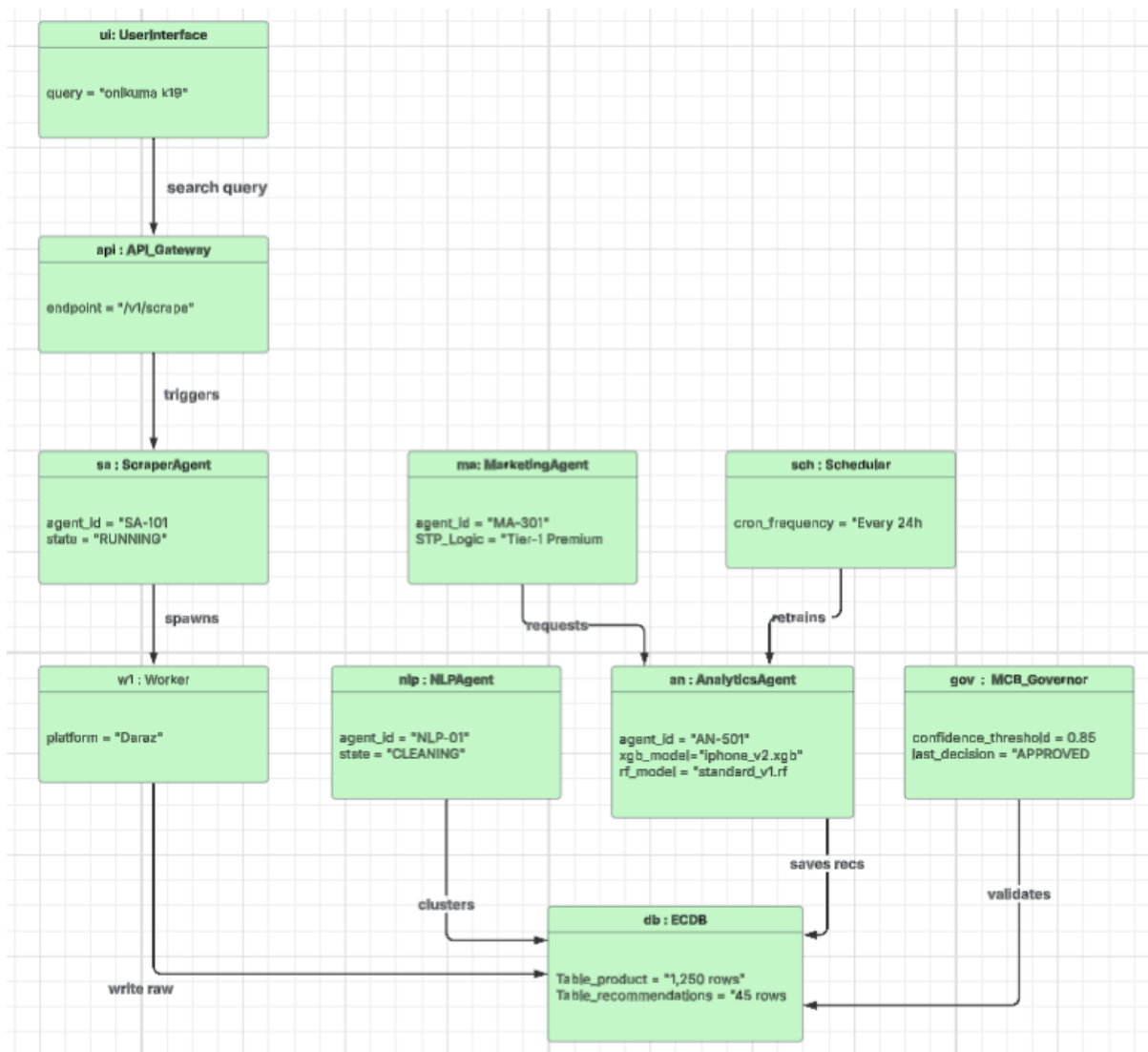


Figure 5.4 Sequence Diagram of NLP Cleaning Sequence

### 5.4.2.3 Sequence Diagram of Strategic Decision Flow



### 5.4.3 Object Diagram



*Figure 5.5 Object diagram of StrategAI*

The user enters a product search, which the API Gateway sends to the Scraper Agent to collect data from platforms like Daraz. The NLP Agent cleans the data, and the Analytics Agent runs machine-learning models to generate insights and recommendations, which are stored in the database. The MCR Governor then validates the results before they're approved.

## 5.5 Software Design Principles

In keeping with the “Pure Project” vision, and to make it actionable over time, StrategAI is built around four governing design principles intended to keep it modular, interpretable, resilient, and technologically state-of-the-art as it evolves. They do not pertain



directly to performance or functionality but rather shape how any agent interacts with the system by setting boundaries ensuring transparency of reasoning and maintaining long-term architectural integrity.

### **Agentic Decoupling**

Every agent in StrategAI is constructed as a completely independent black-box, i.e. the inner workings of an agent, its tools and methods of operations do not affect and constrain another. An example is that the Scraper Agent can be completely replaced - potentially with a better API or reformed with alternative libraries - without any modification to the Marketing Agent or Analytics Agent. This company separation does not only ease the maintenance but it is also possible to consider new ideas because it is not a system that cannot be changed leaving the other parts in an unstable position.

### **Shared Memory Persistence**

Direct agent to agent communication is avoided by design. All the agents interact using only the E-Commerce Database (ECDB) which is seen as shared-memory layer and single point of truth. It cannot be any secret or covert interaction, hence the conflict of synchronization is eliminated and the visibility is kept steady. It pre-invites transparency in all her moves and does it in a documented way such that all other agents would be allowed to work with the same base of verified data.

### **Explainability (XAI)**

StrategAI deals with the intelligible intelligence, as opposed to the black-box automation. Each output of the system has two-dimensional explainability signals Confidence Score and Reasoning Trail surfaced using the MCB (Management Control-Based) assessment loop, such that a user is not only shown what the system is proposing but can also see the reasons why it has made such a recommendation and more importantly its

certainty. This generates trust and instills accountability, audibility that are AI system attributes when it affects business decisions in the real world.

### **Technology Autonomy**

The platform is created to be independent in terms of deployment hence ensuring that the technology stack is not tied to a specific hosting environment. REST-interfaces entirely enable frontend to communicate with backend, therefore implying that StrategAI might be implemented locally on a developer machine, or infinitely scaled on cloud-native systems such as Supabase or Vercel with ease without a rearchitecture redesign. This freedom ensures the system remains flexible, portable, and future-proof due to the dynamism in the needs of the infrastructure across time.

## 5.6 Entity Relationship Diagram

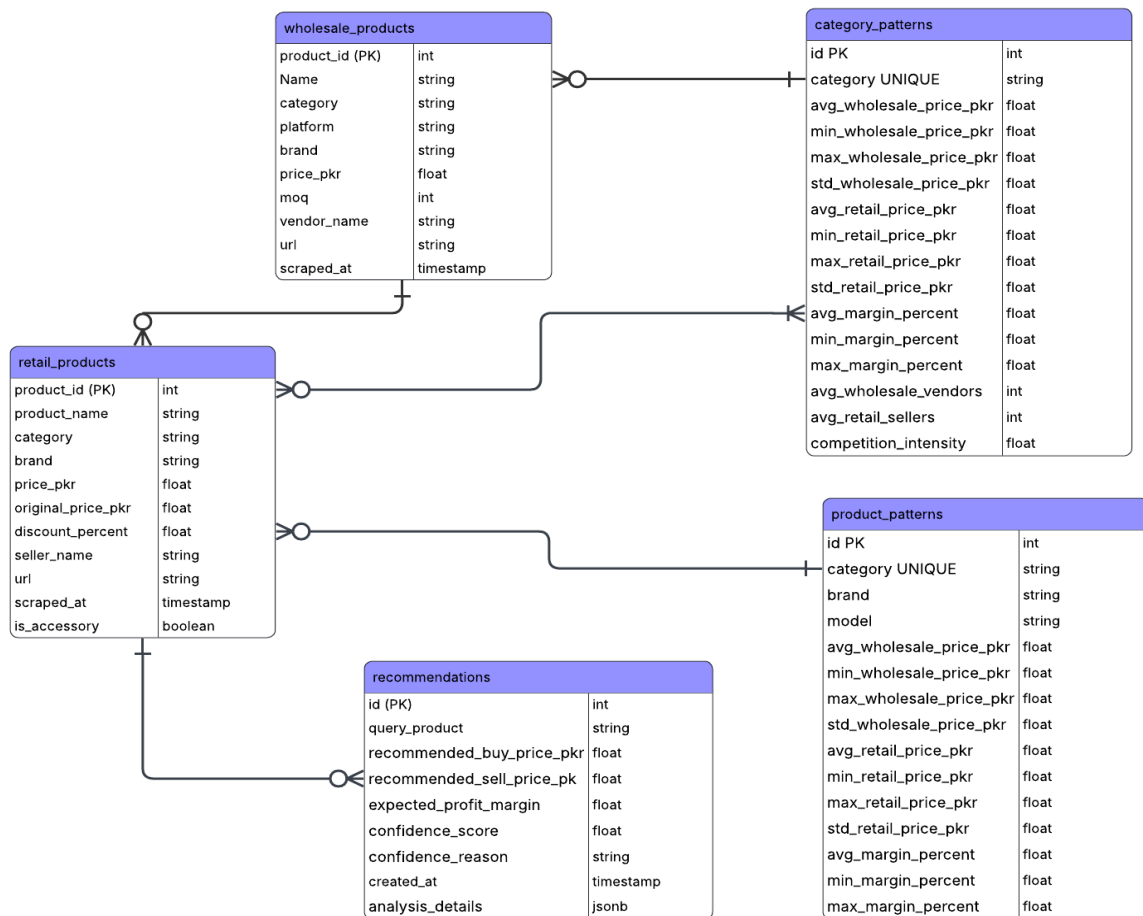
The ERD is designed for **Intelligence Continuity**.

**Input Layer:** Products, Vendors, Platforms.

**Processing Layer:** Clusters, Similarity Scores, Entity Matches.

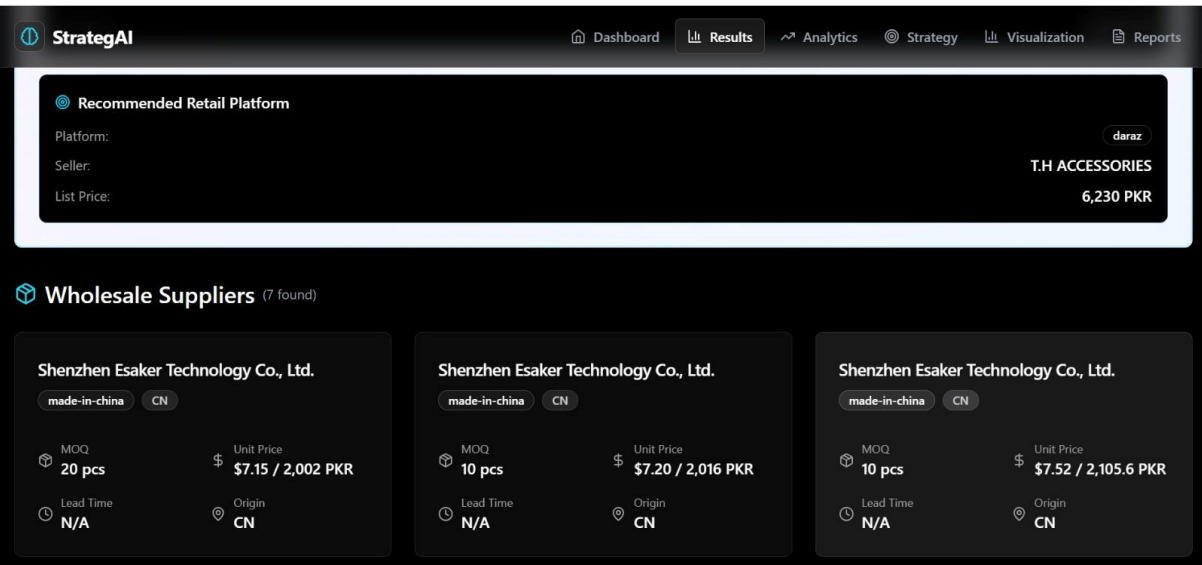
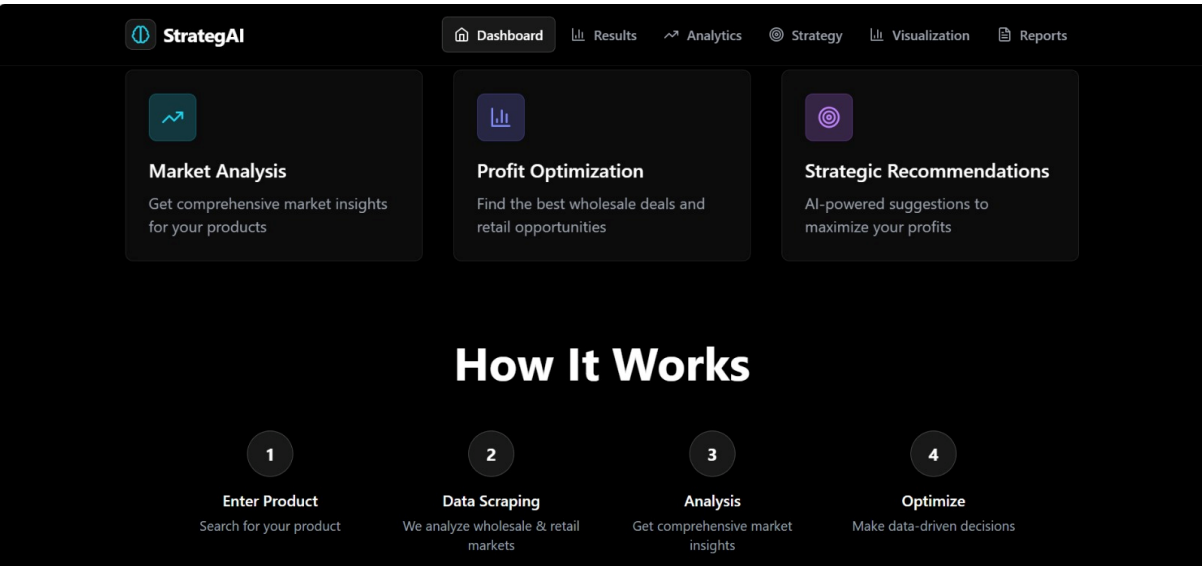
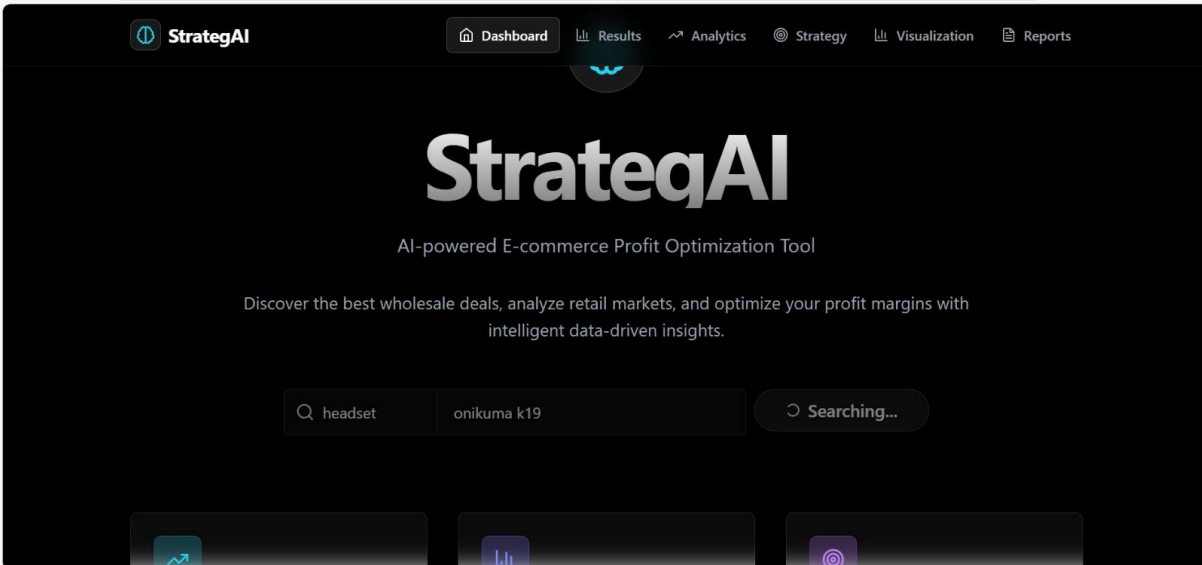
**Intelligence Layer:** Category Patterns, Brand Baselines (Historical Margins).

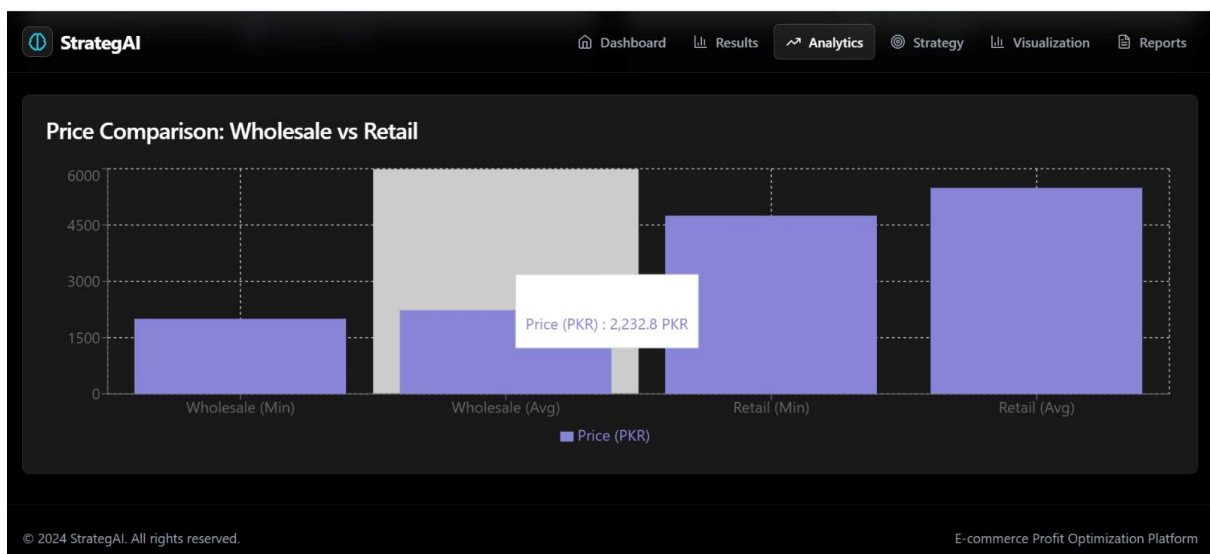
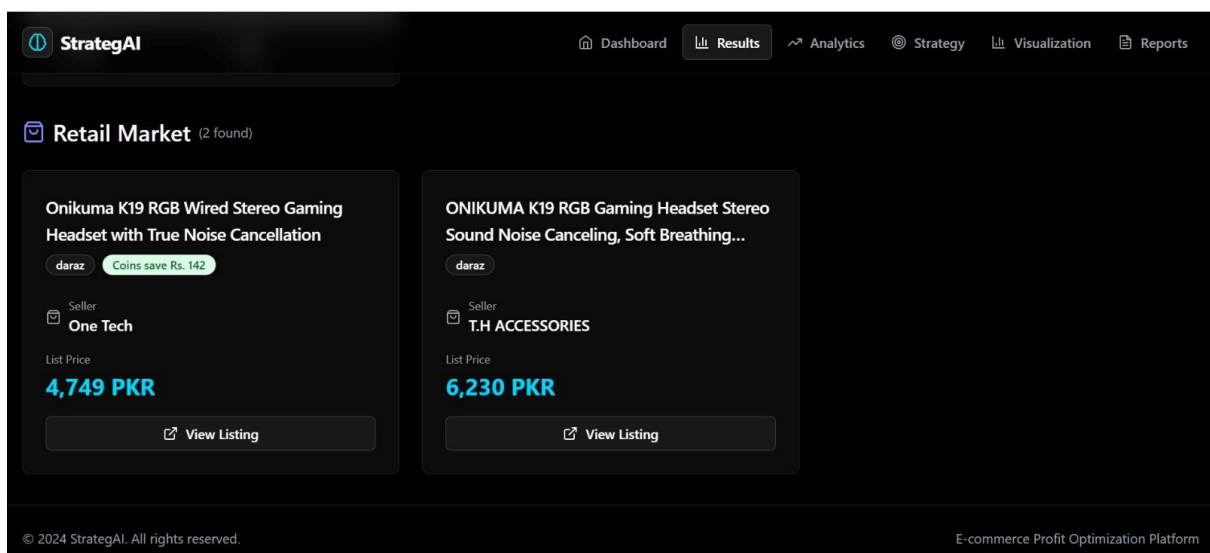
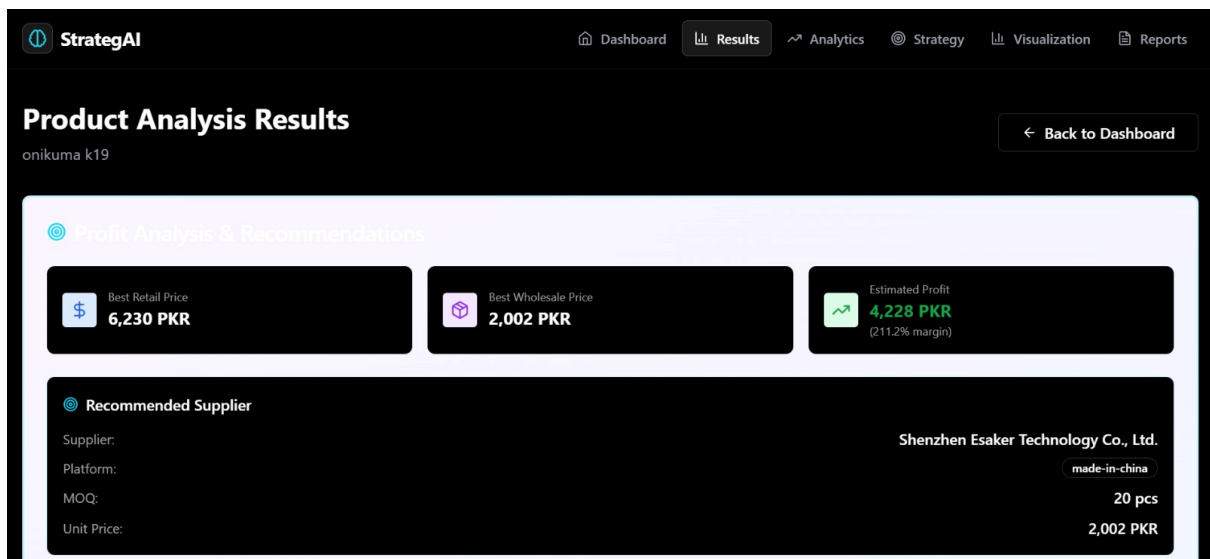
**Action Layer:** Recommendations, Predicted ROI, Ad Campaigns.

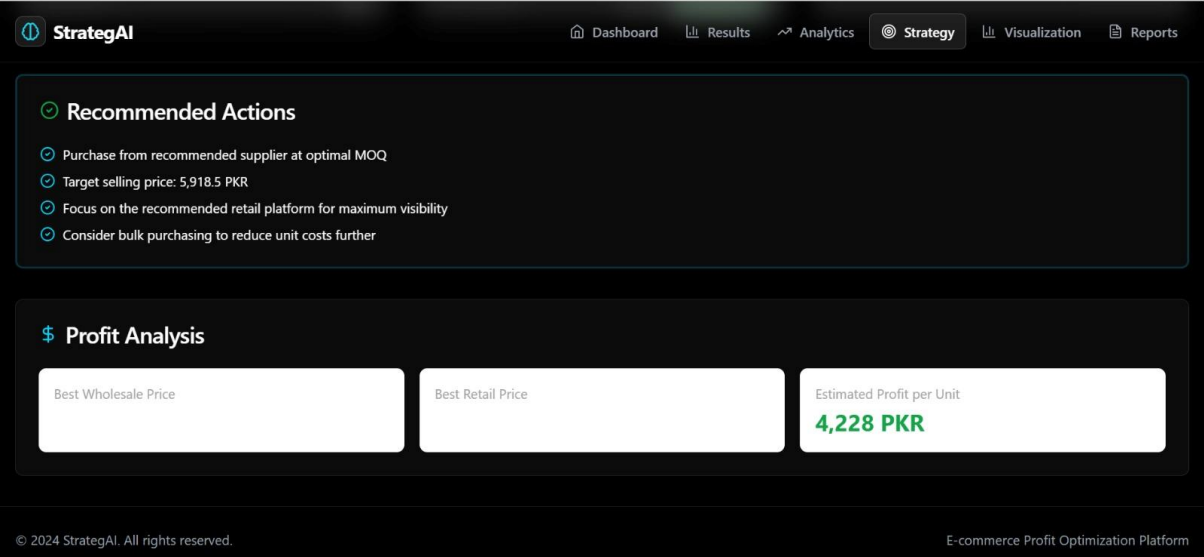
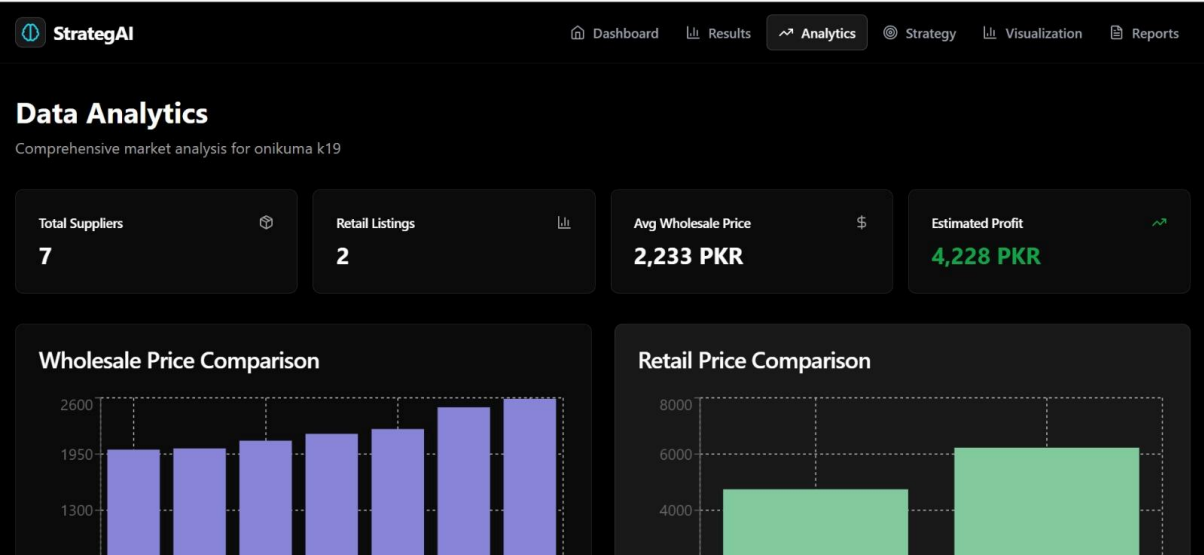


*Figure 5.6 Entity Relationship Diagram of StrategAi*

## 5.7 Interactive User Interface:







StrategAI

DashboardResultsAnalyticsStrategyVisualizationReports

# Sales Strategy

AI-powered recommendations for onikuma k19

Optimal Purchase Strategy

Recommended Supplier:

Shenzhen Esaker Technology Co., Ltd.

Recommended

Minimum Order Quantity:

20 units

Unit Cost:

2,002 PKR

Lead Time:

N/A

Sales Strategy

Target Platform:

daraz

Recommended

Competitive Price Range:

4,749 - 6,230 PKR

Recommended Selling Price:

5,918.5 PKR

Recommended

Profit Margin:

211.2%

Recommended

Market Insights

Total Suppliers Found:

7

Total Retail Competitors:

2

Market Opportunity:

High - Good profit potential

Recommended

StrategAI

DashboardResultsAnalyticsStrategyVisualizationReports

# Product Report

Comprehensive analysis for onikuma k19

Download Report

Executive Summary

Total Suppliers

Retail Listings

Best Wholesale Price

Estimated Profit

4,228 PKR

Wholesale Suppliers Analysis

| Supplier                             | Platform      | Unit Price | MOQ      |
|--------------------------------------|---------------|------------|----------|
| Shenzhen Esaker Technology Co., Ltd. | made-in-china | 2,002 PKR  | 20 units |

StrategAI

DashboardResultsAnalyticsStrategyVisualizationReports

# Retail Market Analysis

|          |                 |            |                                                                      |
|----------|-----------------|------------|----------------------------------------------------------------------|
| Platform | Seller          | List Price | Product Title                                                        |
| daraz    | One Tech        | 4,749 PKR  | Onikuma K19 RGB Wired Stereo Gaming Headset with True Noise...       |
| daraz    | T.H ACCESSORIES | 6,230 PKR  | ONIKUMA K19 RGB Gaming Headset Stereo Sound Noise Canceling, Soft... |

StrategAI

Dashboard
Results
Analytics
Strategy
Visualization
Reports

### Wholesale Suppliers Analysis

|                                      |               |             |          |
|--------------------------------------|---------------|-------------|----------|
| Supplier                             | Platform      | Unit Price  | MOQ      |
| Shenzhen Esaker Technology Co., Ltd. | made-in-china | 2,002 PKR   | 20 units |
| Supplier                             | Platform      | Unit Price  | MOQ      |
| Shenzhen Esaker Technology Co., Ltd. | made-in-china | 2,016 PKR   | 10 units |
| Supplier                             | Platform      | Unit Price  | MOQ      |
| Shenzhen Esaker Technology Co., Ltd. | made-in-china | 2,105.6 PKR | 10 units |
| Supplier                             | Platform      | Unit Price  | MOQ      |
| Shenzhen Esaker Technology Co., Ltd. | made-in-china | 2,184 PKR   | 10 units |
| Supplier                             | Platform      | Unit Price  | MOQ      |
| Shenzhen Esaker Technology Co., Ltd. | made-in-china | 2,240 PKR   | 10 units |



## **6. Algorithm Analysis and Complexity**

The key algorithms in this chapter facilitate StrategAI to operate in an environment whose business aspects continue to vary with tremendous amounts of data, considerable variability, and velocity. It is not based on the basic series scraping or logic of rules but it is based on a hybrid of efficient techniques scalable models that ensures the quality output when it interacts with noisy, unstable or unpredictable sources. All the algorithms are evaluated in the multiple different aspects related to the functionality and the performance and the computational complexity. Such an evaluation will reveal that the overall system will retain its agility, reliability and scalability once more and more data and users are supported without any degeneration.

### **6.1 Modular Sieve Algorithm (Data Acquisition)**

The Scraper Agent runs on the Modular Sieve Algorithm and acts much more like an intelligent harvester continuously feeding cleaned and validated real-time market data than a regular scraper which simply lifts whatever is available on a web page without discrimination. It includes live filtering, normalization, and contextual validation to ensure only the relevant usable product information makes it into ECDB - essential to keep the database clean. Parallel processing merged with structured rule-based screening means fast execution but with a high degree of accuracy maintained because this has to be dynamic due to changing e-commerce data. Also, the algorithm keeps strength against changes of UI or data-format across different e-commerce sites, being able to adjust fast and well to any new and odd things it finds. So, this new plan boosts the whole effect on collecting and handling data in the quickly changing online market place.

### 6.1.1 Pseudocode

```
ALGORITHM ModularSieve(Query, Categories):  
  Initialize ThreadPoolExecutor with P platforms  
  FOR each platform in Platforms DO IN PARALLEL:  
    Fetch raw HTML using Playwright/Requests  
    Extract [Title, Price, Supplier] via CSS Selectors  
  FOR each product in RawResults DO:  
    IF product.price is NULL OR product.title is NULL: DISCARD  
    IF NOT TokenMatch(product.title, Query): DISCARD  
    IF IsAccessory(product.title, Category): DISCARD  
  Normalize pricing to PKR  
  RETURN CleanedList
```

This structure reveals how the algorithm first disperses workload across several platforms at the same time then puts each product through a quality-control pipeline that removes incomplete, irrelevant, or misleading listings before standardizing final outputs.

### 6.1.2 Complexity & Performance

#### Time Complexity:

The primary computational load comes from iterating through the product results. Since the scraping operations are conducted in parallel across platforms, the time complexity essentially becomes,

$$O(N)$$

where N is the total number of product records retrieved, and is largely invariant to the number of platforms P due to concurrency.

#### SpaceComplexity:

Space consumption increases only based on the count of valid (filtered) records kept meaning.

$$O(M)$$

where M is the number of cleaned product entries that pass the sieve. Space for invalid/noisy entries is freed early and does not add unnecessary memory overhead.

## Performance Comparison

It shows how the Modular Sieve differs from, and enhances a standard scraping workflow

| Feature      | Traditional Scraper                               | Modular Sieve (StrategAI)                    |
|--------------|---------------------------------------------------|----------------------------------------------|
| Resilience   | Low (UI changes break logic easily)               | High (Modular, rule-driven filtering)        |
| Speed        | Slow — sequential execution                       | Fast — parallel worker pool                  |
| Data Quality | Noisy — accessories and irrelevant items included | Clean — smart keyword and category filtering |

*Table 6.1 Performance comparison of Modular Sieve Algorithm*

To wrap it up, the Modular Sieve Algorithm boosts not just data accuracy but also makes sure that StrategAI stays scalable, tough against platform tweaks, and able to work right when needed which is crucial for dynamic pricing and competitive analytics.

## 6.2 NLP Clustering (TF-IDF + Cosine)

The Clustering Algorithm is a process that will be performed centrally in the NLP Agent, it will bring together the separate product entries in a cohesive solid structure called the Master Entity. Naive matching schemes do not tend to work since vendors almost always call the same product by another name, abbreviation, or using a brand name. StrategAI overcomes this shortcoming by TF-IDF vectorization and cosine similarity - recognition of semantic proximity as opposed to matching literal text. In a nutshell, it allows the platform to establish set listings that are stated in various forms yet, have the same meaning. This is extremely essential to the functionality of analytics as well as pricing strategy or any decision intelligence.

### 6.2.1 Pseudocode

ALGORITHM NLP\_Alignment(ProductList):

Clean text (Remove punctuation, lowercase, stop-word removal)

Vectorizer = TfidfVectorizer()

```
Matrix = Vectorizer.fit_transform(ProductList.titles)
```

```
Similarity_Matrix = Calculate_Cosine_Similarity(Matrix)
```

Cluster products where Similarity > 0.85

Assign unique ClusterID to each group

Store CleanedResults in ECDB

This flow first tidies up text, then turns every product title into a number-based TF-IDF vector form. By finding the cosine similarity matrix, the algorithm checks how much two titles look like each other in vector space. Products that meet the similarity mark (for example, 0.85 or more) are put into groups and given a common ClusterID, which is saved back into the ECDB for later use.

## 6.2.2 Complexity & Performance

### Time Complexity

The computational cost comes mostly from two places:

- Building the TF-IDF matrix over the whole vocabulary
- Calculating cosine similarity over all product pairs

This gives a complexity of:

$$O(V \times N^2)$$

where V is the vocabulary size and N is the number of product records. It is quadratic with respect to dataset size. However, the resulting semantic alignment accuracy is much improved for downstream intelligence.

| Algorithm            | Accuracy | Compute Cost | Reasoning                                                     |
|----------------------|----------|--------------|---------------------------------------------------------------|
| Regex Match          | 40%      | Very Low     | Breaks when word order or phrasing changes                    |
| Levenshtein Distance | 65%      | Medium       | Captures character edits but scales poorly for large datasets |
| TF-IDF (StrategAI)   | 88%      | Medium       | Best captures semantic similarity across vendors              |

*Table 6.2 Performance comparison of TF-IDF + Cosine*

## Performance Comparison

This table below sheds light on how TF-IDF + cosine similarity fares when pitted against simpler approaches that are based on matching text:

Though TF-IDF does require more computation compared to simple string matching, it affords significantly higher alignment accuracy thereby improving such use cases as pricing intelligence, duplicate detection, and market-level insights. This makes it an optimal choice considering the balance between computational efficiency and semantic precision for StrategAI's NLP layer.

## 6.3 Ensemble Price Prediction

The Analytics Agent is powered by the Ensemble Price Prediction Algorithm, which drives the final best price recommendation that gets delivered by StrategAI to this business user. While most solutions run with a single predictive model, StrategAI leverages the power of XGBoost and Random Forest hybrid ensemble approaches toward ensuring that it learns both the complex nonlinear relations and generalized trend stability within the market. Therefore, based on weighting outputs from each model and intelligently blending them together, this algorithm will result in guidance pertaining to prices that shall not only be accurate but also resilient to noise across markets, imbalanced data, and sudden fluctuations from competitors.

This ensemble pipeline is a part of real-world business logic — taking into account such factors as wholesale price trends, competitive pressure, and retail price boundaries. What comes out at the other end is not just a mathematical value in isolation but rather an answer to how to optimize profit while staying competitive and controlling risk.

### 6.3.1 Pseudocode

```
ALGORITHM EnsemblePrediction(MarketFeatures):  
    Load Pre-trained XGBoost and Random Forest Models  
    Feature Engineering: [Wholesale_Avg, Retail_Min, Competition_Count]  
    Pred_XGB = XGB_Model.predict(Features)  
    Pred_RF = RF_Model.predict(Features)  
    Final_Price = (0.6 * Pred_XGB) + (0.4 * Pred_RF)  
    Calculate Confidence = RMSE_Weightage / Variance  
    RETURN Final_Price, Confidence
```

Key market indicators are first extracted and transformed into structured features. The models already trained on historical ECDB data, generate prices independently of each other. StrategAI then returns the final recommendation as a weighted average, with slightly more weight from XGBoost since it is better at capturing nonlinearity in behavior. In parallel, variance-based confidence is calibrated with RMSE so that every price recommendation carries an interpretable confidence signal for the MCB Decision Agent and end user.

### 6.3.2 Complexity & Performance

#### Inference Complexity

In prediction — i.e., inference — the computational cost depends on the number of trees and their depths in each ensemble model used, such that it is proportional to the product of the number of trees and tree depth.

$$O(Trees \times Depth)$$

As both XGBoost and Random Forest are optimized implementations towards fast inference, this complexity is still very much within the real-time operational constraints even for big batches of items, making this architecture very apt for live dynamic-pricing environments.

## Performance Comparison

The following comparison shows how the ensemble approach performs when it is compared to commonly used approaches:

| Model                | RMSE (Error) | Stability | Explainability |
|----------------------|--------------|-----------|----------------|
| Linear Regression    | High         | High      | High           |
| XGBoost              | Low          | Medium    | Medium         |
| Ensemble (StrategAI) | Very Low     | High      | High           |

*Table 6.3 Performance prediction of Ensemble Price Prediction*

Linear regression stays basic and simple to read though it fails at handling nonlinear dynamics in pricing. XGBoost gives very high accuracy but may suffer from sensitivity to changing data conditions. Hence, by combining both ensemble learners, StrategAI attains the best balance between low prediction error and robustness under market variability together with explaining model behavior.

Thus, the Ensemble Price Prediction Algorithm makes the pricing intelligence of StrategAI accurate, resilient, and business-aware to enable organizations to implement confident as well as data-driven price strategies in competitive e-commerce environments.

## 6.4 Strategic Marketing Logic

The Strategic Marketing Logic Algorithm is the intelligence core of the Marketing Agent; hence, it guides how StrategAI transforms analytical insights into actionable business strategy. The pricing engine only deals with numbers and optimization; however, the Marketing Agent is responsible for positioning, messaging, and channel allocation based on STP which stands for classic Segmentation Targeting Positioning framework.

The algorithm has not been set up with a view of automating mere marketing tasks but rather one that tries to replicate in some degree of human CMO strategic reasoning by using margins,

product category signals, and conversion forecasts to determine the most profitable go-to-market approach. Over time, this logic will keep evolving even further toward higher forms of behavior-adaptive learning, persona clustering, response-based campaign iteration-the foundation henceforward-would therefore be self-optimizing marketing intelligence.

#### **6.4.1 Pseudocode (STP Logic)**

ALGORITHM MarketingLogic(ProductData, AnalyticsData):

Analyze Margins: IF Margin > 30% THEN Segment = "Premium" ELSE "Budget"

Target: IF Category in ['Tech', 'Gadgets'] THEN Channel = "Google Search"

Promote: Create AdCopy based on [Product\_Name + " Best Price in Pakistan"]

Forecast:  $ROI = (Estimated\_Conversion * Margin) - Ad\_Spend$

RETURN Marketing\_Plan\_Object

The algorithm finds out which segment of the market the product occupies at the first stage by using the margin strength as a proxy in price, which is not the price and the net profitability. In case the general combination is more advantageous it will be under Premium. In case some products pull down margins, the lower products come under this category of Budget. Product tone and positioning is thus affected. Then, it selects the optimal channel, such as tech products can perform well in the intent led curriculum platforms like Google Search. The system automatically develops persuasive advertisement texts touching upon product name and price dependent on the geo location so obvious and close to the purchaser in his or her district. Finally, a forecasted ROI would be derived by linking the likelihood of conversion occurring and the margin and expense that is going to be incurred. This allows the StrategAI to give the financially sound strategies as opposed to the guesses.



### 6.4.2 Performance Comparison

The following comparison depicts how automatic strategic marketing logic works better and more consistently than manual planning:

| <b>Manual Planning</b>                        | <b>StrategAI Agent</b>                                     |
|-----------------------------------------------|------------------------------------------------------------|
| Takes hours of market research                | Takes < <b>5 seconds</b> of computation                    |
| Decisions influenced by bias or intuition     | Decisions are <b>data-driven and margin-backed</b>         |
| Strategy varies widely between individuals    | Strategy is <b>consistent, explainable, and repeatable</b> |
| Scaling to many products requires large teams | <b>Scales instantly</b> across entire catalogs             |

*Table 6.4 Performance comparison of Strategic Marketing Logic*

It replaces subjective judgement with structured analytical reasoning ensuring alignment of the marketing strategies to profitability, competitive positioning and measurable ROI-with reduced effort and decision-time cut dramatically.

## 7. Appendices

### 7.1 Glossary of Terms

| Term                | Definition                                                                                                                 |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Agentic AI</b>   | Artificial Intelligence designed to act as an agent, capable of performing tasks autonomously to achieve specific goals.   |
| <b>API</b>          | Application Programming Interface - The set of protocols used for building and integrating application software (FastAPI). |
| <b>CrewAI</b>       | A framework for orchestrating role-playing, autonomous AI agents.                                                          |
| <b>LangGraph</b>    | A library for building stateful, multi-actor applications with LLMs, used for the project's core workflow.                 |
| <b>MAS</b>          | Multi-Agent System - A computerized system composed of multiple interacting intelligent agents.                            |
| <b>Next.js</b>      | A React framework for building full-stack web applications with optimized performance.                                     |
| <b>Scraping</b>     | The process of automatically extracting data from websites (implemented via Playwright and BS4).                           |
| <b>Tailwind CSS</b> | A utility-first CSS framework used for the project's premium dark mode styling.                                            |

### 7.2 Annexes

#### Annex B: Organizational Context

- **Appendix B-1: About StrategAI Initiatives:** StrategAI is developed as a tool to bridge the gap between large-scale market data and small-scale retail operations.
- **Appendix B-2: Target Organization Summary:** (2-3 pages regarding the business unit or incubation entity hosting the project). Focus on the "AI for All" mission and the departmental hierarchy.

#### Annex C: Research and Background Evidence

- **Appendix C-1: Core Research Paper:** *"Orchestrating Autonomous Agents for Market Sentiment Analysis"* – This research served as the technical foundation for the CrewAI/LangGraph integration.
- **Appendix C-2: Background Analysis:** Detailed analysis of competitive landscapes

in e-commerce, identifying the need for a unified retail-wholesale arbitrage platform.

- **Appendix C-3: UML Architecture:**

- **Sequence Diagram:** Detailing the flow from user search -> Scraper Agent -> NLP Processing -> Analytics Output.
- **Component Diagram:** Showing the decoupled relationship between the FastAPI backend and Next.js frontend.

## **Annex D: Hardware, Software Analysis and Requirement**

- **Appendix D-1: Hardware Specifications:**

- **Local Development:** 16 vCPUs, 32GB RAM, NVIDIA T4 GPU for local model inference.
- **Cloud Hosting:** AWS EC2 instance for backend services and Vercel for the Next.js frontend.

- **Appendix D-2: Software Analysis (Tech Stack):**

- **Backend:** Python, FastAPI, LangGraph, CrewAI, BeautifulSoup4, Playwright, **Streamlit (Internal Developer Dashboard)**.
- **Frontend:** Next.js (App Router), TypeScript, Tailwind CSS, Recharts (for user dashboard visualization), Framer Motion.
- **Database:** Supabase / PostgreSQL.

- **Appendix D-3: Use Case Narrations:**

- **UC-01: Price Comparison Search:** User enters a product name; system triggers scraping and analysis; results are displayed in comparison cards.
- **UC-02: Strategy Generation:** System analyzes market trends and provides pricing strategies for sellers.
- **UC-03: Developer Debugging:** Developers use the Streamlit dashboard to monitor agent execution logs and visualize intermediate data states.

## **Annex E: Software Design and Modeling**

- **Appendix E-1: Database Schema:** Full ERD showing relations

between SearchHistory, MarketData, AgentLogs, and UserPreferences.

- **Appendix E-2: Detailed UI Design:** Style guide for the "Glassmorphism" theme, including color variables (--primary-cyan, --secondary-indigo) and standard component variants (Cards, Buttons).

- **Appendix E-3: Data Visualization Architecture:**

- **Frontend (User):** Usage of **Recharts** to render dynamic market trends,

histograms, and profit comparisons in the main dashboard.

- **Internal (Developer):** Implementation of a **Streamlit** dashboard for real-time monitoring of CrewAI agent orchestration and data cleaning proofs.

#### **Annex F:** Software Algorithm and Complexity

- **Appendix F-1: Core Orchestration Algorithm:**

Implementation details extracted from `complete_workflow.py`.

Workflow Logic Overview:

1. Initialize StateGraph with AgentState.
2. Define nodes: 'scraper', 'nlp', 'analytics'.
3. Define conditional edges: 'should\_clean' and 'should\_analyze'.
4. Compile the graph to create the executable application.

- **Appendix F-2: Time Complexity Analysis:**

- **Data Acquisition:**  $O(P * S)$  where P is platforms and S is search depth.
- **Agent Reasoning:**  $O(K)$  where K is a constant factor of LLM context window limits.

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